CHAPTER 6

BUILDING SERVICES AND SYSTEMS

General Comments

This chapter focuses on building systems and services, potential related safety hazards and when and how they should be installed. In some cases, many of the provisions are located in other portions of the code. This chapter brings together all building system- and service-related issues for convenience and provides a more systematic view of buildings. The following building services and systems are addressed:

- Electrical equipment, wiring and hazards, including solar photovoltaic power systems (Section 603).
- Elevator recall and maintenance (Section 604).
- Fuel-fired appliances (Section 605).
- Commercial cooking equipment and systems (Section 606).
- Commercial cooking oil storage tank systems (Section 607).
- Mechanical refrigeration (Section 608).
- Hyperbaric facilities (Section 609)
- Clothes dryer exhaust systems (Section 610).

Some of the sections specifically deal with installation while others deal with reducing the hazards from the use of the services or systems. For example, Section 603 notes that using too many extension cords on the building electrical system may present a fire hazard. On the other hand, the discussion of elevator recall and maintenance in Section 604 simply states when and how recall is required.

Purpose

As technology progresses and societal expectations increase, building systems and services become more complex and numerous. This chapter is intended to direct the user and inspector to the correct installation, maintenance and inspection requirements applicable to building systems and appliances. The requirements make the appropriate cross references to the *International Fuel Gas Code*® (IFGC®), *International Mechanical Code*® (IMC®) and *National Electrical Code* (NFPA 70), as applicable.

SECTION 601—GENERAL

601.1 Scope. The provisions of this chapter shall apply to the installation, operation, testing and maintenance of the following building services and systems:

1. Electrical systems, equipment and wiring.
2. Information technology server rooms.
3. Elevator systems, emergency operation and recall.
4. Fuel-fired appliances, heating systems, chimneys and fuel oil storage.
5. Commercial cooking equipment and systems.
6. Commercial cooking oil storage.
7. Mechanical refrigeration systems.
8. Hyperbaric facilities.
9. Clothes dryer exhaust systems.

C This section establishes the applicability of the chapter to a variety of building systems and services when they are being installed, during their operation and for long-term maintenance. Note that the provisions related to energy, including standby and emergency power, photovoltaic installations and energy storage systems (batteries), are located in Chapter 12.

601.2 Hazard abatement. Operations or conditions deemed unsafe or hazardous by the *fire code official* shall be abated. Equipment, appliances, materials and systems that are modified or damaged and constitute an electrical shock or fire hazard shall not be used.

C This section addresses general abatement of hazards and unsafe conditions for all building services and systems. This section was created based on the concepts in Section 603.2. A mechanism is needed on a general level to be able to address hazards and unsafe conditions.

601.2.1 Correction of unsafe conditions. The *fire code official* shall be authorized to require the *owner*, the *owner's* authorized agent, operator or occupant of a building or premises to abate or cause to be abated or corrected such unsafe operations or conditions either by repair, rehabilitation, demolition or other *approved* corrective action in compliance with this code.

C The purpose of this section is to establish the fire code official's authority, identify the responsible person(s) who will be addressing the correction of the unsafe conditions, and establish what criteria are to be followed for the corrections. Section 601.2.1 is based on Section 115.6.

SECTION 602—DEFINITIONS

602.1 Definitions. The following terms are defined in Chapter 2:

COMMERCIAL COOKING APPLIANCES.

HOOD.

Type I.

REFRIGERANT.

REFRIGERATING (REFRIGERATION) SYSTEM.

C Definitions of terms can help in the understanding and application of the code requirements. This section directs the code user to Chapter 2 for the proper application of the indicated terms used in this chapter. Terms may be defined in Chapter 2, in another International Code® (I-Code®) as indicated in Section 201.3, or the dictionary meaning may be all that is needed (see also commentary, Sections 201.1 through 201.4).

SECTION 603—ELECTRICAL EQUIPMENT, WIRING AND HAZARDS

603.1 General. Electrical equipment, wiring and systems required by this code or the *International Building Code* shall be installed, used and maintained in accordance with NFPA 70 and Sections 603.2 through 603.9.

C This section provides scoping clarification that both NFPA 70 and this section apply to the installation, use and maintenance of electrical equipment, wiring and systems required by both this code and *International Building Code*® (IBC®).

603.1.1 Equipment and wiring. All electrical equipment, wiring, devices and appliances shall be tested; *listed* and *labeled;* and installed, used and maintained in accordance with NFPA 70 and all instructions included as part of such listing.

C The fire code official should look for the listing mark of an approved testing or inspection agency on equipment, wiring, devices and appliances. In some cases, the fire code official may need to request the agency's published report showing the listing to verify that the equipment, wiring, devices and appliances meet the requirements of NFPA 70. Fire code officials experiencing difficulties interpreting the marking or listing of an agency, as in the case of a laboratory not located within the United States, should consult representatives of the agency or the US Consumer Products Safety Commission (CPSC) for assistance.

603.1.2 Healthcare facilities. In Group I-2 facilities, ambulatory care facilities and outpatient clinics, the electrical systems and equipment shall be maintained and tested in accordance with NFPA 99.

C In order to meet federal conditions of participation, the electrical systems and equipment of health care facilities must be maintained and tested according to the requirements listed in NFPA 99, Section K913. This section also aligns with electrical systems maintenance and testing requirements for outpatient clinics, Group B ambulatory care and Group I-2 facilities.

603.2 Abatement of unsafe conditions and electrical hazards. Conditions that constitute an electrical shock or fire hazard shall be abated.

C The purpose of this section is to require that any conditions that create an electrical shock or fire hazard need to be addressed.

603.2.1 Modified or damaged. Electrical wiring, devices, equipment and appliances that are modified or damaged, and constitute an electrical shock or fire hazard, shall not be used until repaired or replaced in accordance with this code and NFPA 70.

C Maintenance of electrical systems and services to achieve compliance with the requirements of NFPA 70 is required. The leading causes of electrical fires include inadequate or improper maintenance, nonconforming modifications to existing installations, failure to maintain clearances around electrical equipment and devices, and improper use of electrical equipment and devices. A detailed analysis of the causes of residential electrical fires by the US Consumer Products Safety Commission suggests that misuse and improper modifications to conforming electrical systems are the leading causes of these fires.

603.2.2 Open electrical terminations. Open junction boxes and open-wiring splices shall be prohibited. *Approved* covers shall be provided for all switch and electrical outlet boxes.

C Without covers, connections made in junction boxes may be subject to physical damage. Such damage may loosen electrical connections, resulting in high-resistance arcing. Switches and outlet boxes are subject to arcing from loose connections and reduced clearances between contacts as they age. Accumulation of dirt and debris in open electrical boxes creates an ignitable fuel concentration. Fires in open electrical boxes may spread to wire or cable insulation or other fuels in electrical and mechanical concealed spaces. Furthermore, unprotected electrical connections are electrical shock hazards to personnel working in concealed spaces.

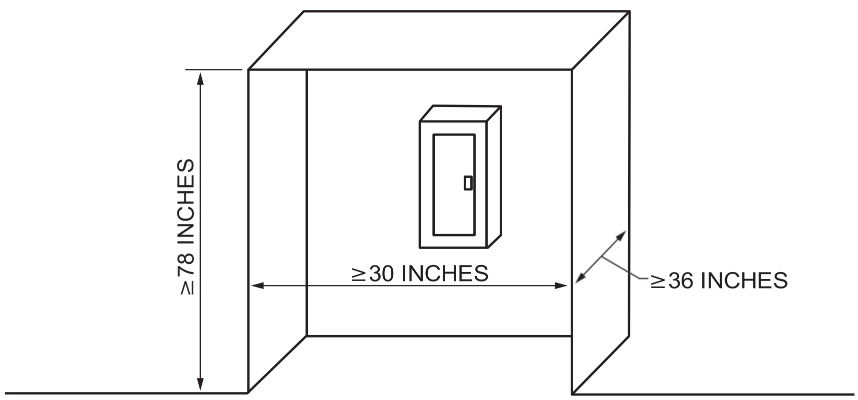
603.3 Illumination. Illumination shall be provided for service equipment areas, motor control centers and electrical panelboards.

C Adequate lighting in electrical service distribution equipment areas is required to facilitate the location of the electrical service shutoff during a fire or other emergency and to minimize potential hazards during maintenance or repair work. Although not required, this lighting should be connected to an emergency or standby power source to permit continued illumination when power to service equipment is interrupted during maintenance repair activities or emergencies.

603.4 Working space and clearances. Working space around electrical equipment shall be provided in accordance with Section 110.26 of NFPA 70 for electrical equipment rated 1,000 volts or less, and Section 110.32 of NFPA 70 for electrical equipment rated over 1,000 volts. The minimum required working space shall be not less than 30 inches (762 mm) in width, 36 inches (914 mm) in depth and 78 inches (1981 mm) in height in front of electrical service equipment. Where the electrical service equipment is wider than 30 inches (762 mm), the minimum working space shall be not less than the width of the equipment. Storage of materials shall not be located within the designated working space.

C Adequate clearance serves two important purposes: physical separation of combustibles from heat-producing electrical devices and equipment to minimize the possibility of ignition, and providing adequate space to perform maintenance and repair work safely (see Commentary Figure 603.4). Reference to the appropriate sections of NFPA 70 based on voltage is provided to ensure the clearances are consistent with the standard.

Commentary Figure 603.4—Required Clearances



For SI: 1 inch = 25.4 mm.

603.4.1 Electrical room marking. Doors into electrical control panel rooms shall be marked with a plainly visible and legible sign stating "ELECTRICAL ROOM" or similar *approved* wording.

C In addition to the illumination required in Section 603.3, additional labeling is required for the electrical equipment to assist emergency responders in identifying and shutting down electrical service controls during a fire or other emergency. See also the commentary to rapid shutdown of PV systems in Section 1205.4.

603.4.2 Disconnect means marking. The disconnecting means for each service, feeder or branch circuit originating on a switchboard or panelboard shall be legibly and durably marked to indicate its purpose unless such purpose is clearly evident.

C By requiring each service to be identified, first responders will be warned that disconnecting one service does not necessarily deenergize an entire building or equipment. It will also be clear specifically what system they are disconnecting.

603.4.3 Multiple supply connections marking. Where buildings or structures are supplied by more than one power source, markings shall be provided at each service equipment location and at all interconnected electric power production sources identifying all electric power sources at the premises in accordance with NFPA 70.

C This section recognizes that multiple power sources may be available and in use in a building. The rapid growth in solar energy, energy storage and other alternative power sources means buildings may have more than one power source supporting their electrical needs. This is becoming a more common hazard to first responders. It is important to know where the building or structure is supplied by alternate energy sources. Without this marking, those simply turning off the main disconnect may assume that the building or structure has been fully deenergized. However, there may still be energized systems that pose a serious hazard to first responders.

603.5 Relocatable power taps and current taps. The construction and use of current taps and relocatable power taps shall be in accordance with NFPA 70 and this code.

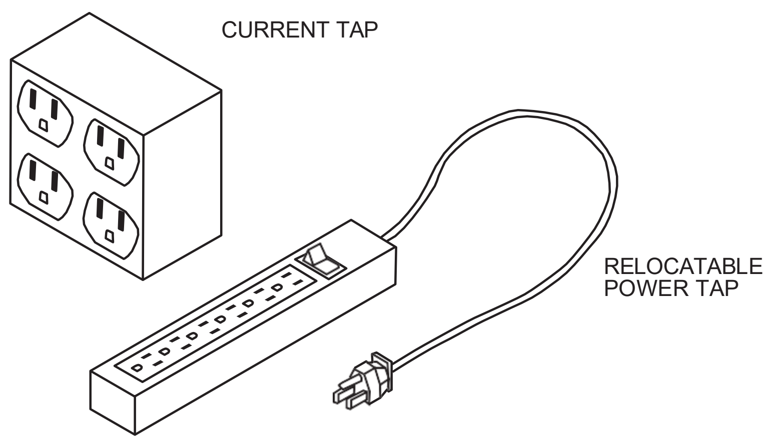
C This section is intended to prohibit conditions that could lead to the overloading of building electrical circuits that, in turn, could result in a fire. The devices intended to be regulated by this section include current taps (sometimes referred to as multiplug adapters) and relocatable power taps (sometimes referred to as power strips).

Overcurrent protection interrupts power to an outlet only when connected loads exceed the current rating of the overcurrent device for a specified amount of time. Where relocatable power taps (RPTs) and current taps are used for several appliances, such devices may produce enough heat to ignite nearby combustibles in the time it takes to trip the overcurrent protection device. Simultaneous operation of many small loads may cause dangerous localized resistance heating without tripping the overcurrent protection device. Additionally, current taps may result in loose electrical connections because of the weight of the cords pulling on them.

The use of these devices may also indicate that the building's electrical wiring is inadequate for the connected loads or occupancy demands. These devices are intended for temporary use only, not at a fixed location or in place of wiring complying with NFPA 70.

The code does allow for the use of listed, relocatable power taps complying with specific criteria mentioned in Sections 603.5.1 through 603.5.3. Commentary Figure 603.5 shows the difference between a current tap and a power tap.

Commentary Figure 603.5—Typical Current Tap and Relocatable Power Tap



603.5.1 Listing. Relocatable power taps shall be *listed* and *labeled* in accordance with UL 1363. Relocatable power taps attached to furnishings shall be listed and labeled in accordance with UL 962A. Current taps shall be *listed* and *labeled* in accordance with UL 498A.

C This section sets out the basic listing requirements for RPTs and current taps.

RPTs are often termed power strips. The testing requirements of UL 1363 are applied to relocatable, cord-connected power taps rated at 250 volts AC or less and 20 amps or less. These devices are intended only for indoor use as a temporary extension of a branch circuit for general use to supply home workshop tools, computers, audio and video equipment, etc. They consist of an attachment plug and a length of flexible cord terminating in an enclosure containing one or more receptacles and may also be provided with fuses or other supplemental overcurrent protection, switches, line surge suppressors or indicator lights. The flexible cord length is dependent on the listing of the particular device, but RPTs have been listed for lengths up to 25 feet (7620 mm). The scope of UL 1363 indicates that RPTs are neither intended to be connected in series nor used with medical equipment.

Current taps are often termed multiplug adaptors. The testing requirements of UL 498A are applicable to current taps. UL 498A is limited to current taps rated to 200A and 600V. UL defines "current tap" as follows: "A male and female contact device that, when connected to an outlet receptacle provides multiple outlet configurations." An adapter, on the other hand, is defined there as "a device that adapts one blade or slot configuration to another (including a grounding adapter for a non-grounding receptacle)." Essentially, current taps allow the use of a single outlet for multiple connections.

Section 603.5.2 addresses the use of RPTs and current taps.

603.5.1.1 Listing in Group I-2 occupancies and ambulatory care facilities. In Group I-2 occupancies and ambulatory care facilities, relocatable power taps shall be *listed* and *labeled* in accordance with UL 1363 except under the following conditions:

1. In Group I-2, Condition 2 occupancies, relocatable power taps providing power to patient care-related electrical equipment in the patient care vicinity, as defined by NFPA 99, shall be *listed* and *labeled* in accordance with UL 1363A or UL 60601-1.

2. In Group I-2, Condition 1 facilities, in care recipient rooms using line-operated patient care-related electrical equipment, relocatable power taps in the patient care vicinity, as defined by NFPA 99, shall be *listed* and *labeled* in accordance with UL 1363A or UL 60601-1.
3. In ambulatory care facilities, relocatable power taps providing power to patient care-related electrical equipment in the patient care vicinity, as defined by NFPA 99, shall be *listed* and *labeled* in accordance with UL 1363A or UL 60601-1.

C Due to the potential for electrical shock to care recipients, the need for a stricter standard for RPTs within the patient care vicinity and resident rooms using line-operating patient care-related electrical equipment is needed. Patient care vicinity is defined in NFPA 99 as an area that is a 6-foot space around a care recipient exam or treatment location. Essentially, where RPTs are being used within the defined 6-foot area around a care recipient, whether in a nursing home, hospital or ambulatory care facility, they must comply with UL 1363A or UL 60601-1. UL 1363A, like UL 1363, is limited to 250 VAC or less but is specifically intended to be used with medical equipment such as monitors or other equipment that is pedestal mounted. Note that general patient care areas or critical patient care areas are defined in NFPA 517 for healthcare facilities. These RPTs contain hospital-grade attachment plugs and hospital-grade outlets.

603.5.2 Application and use. Relocatable power taps and current taps shall be directly connected to a permanently installed receptacle.

Exceptions:

1. Where *approved* for use in a Group A occupancy or in a meeting room in a Group B occupancy, not more than five relocatable power taps shall be permitted to be connected together or connected to an extension cord for temporary use to supply power to electronic equipment.
2. Current taps and relocatable power taps shall not be required to connect directly to a permanently installed receptacle outlet where used for 90 days or less for the purpose of testing the performance of such devices.

C The restrictions on RPTs are similar to those on extension cords. RPTs and current taps should not be used as a substitute for building wiring. RPTs and current taps are intended to be plugged directly into a permanently installed receptacle. RPTs are not intended to be connected in series or connected to other RPTs or extension cords. They are also not intended for use outdoors, at construction sites and at similar hard-use locations.

The exceptions recognize a common acceptable practice for use of such RPTs and also to allow proper testing of RPTs and current taps.

603.5.3 Installation. Relocatable power tap cords shall not extend through walls, ceilings, floors, under doors or floor coverings, or be subject to environmental or physical damage.

C To prevent use as a substitute for permanent wiring, power taps cannot be placed in locations such as within or through walls, under doors or on building surfaces, furnishings, cabinets or similar structures where they would be subject to physical damage. This section would prohibit relocatable power taps from being plugged into a receptacle in one room to power a device in another room.

603.6 Extension cords. Extension cords shall not be a substitute for permanent wiring and shall be *listed* and *labeled* in accordance with UL 817. Extension cords shall not be affixed to structures, extended through walls, ceilings or floors, or under doors or floor coverings, nor shall such cords be subject to environmental damage or physical impact. Extension cords shall be used only with portable appliances. Extension cords marked for indoor use shall not be used outdoors.

C Frequent or improper use of extension cords in place of permanent fixed wiring is another indication of inadequate electrical wiring capacity or incompatible demands (see Commentary Figure 603.6). Physical damage to extension cords caused by concealment or inadequate maintenance may result in localized resistance heating, shorts or ground faults.

Extension cords must also meet the listing and labeling requirements of UL 817. This enables a fire code official to remove inappropriate or nonlisted extension cords. This provides more specific guidance to inspectors and will decrease fire risk.

The amount of electrical current that any extension cord can safely conduct is limited by the size of its conductor, its insulation type and its environment. This principle is often not understood by the general public. As a result, extension cords are commonly overloaded by connecting appliances and other loads in excess of the cord's capacity.

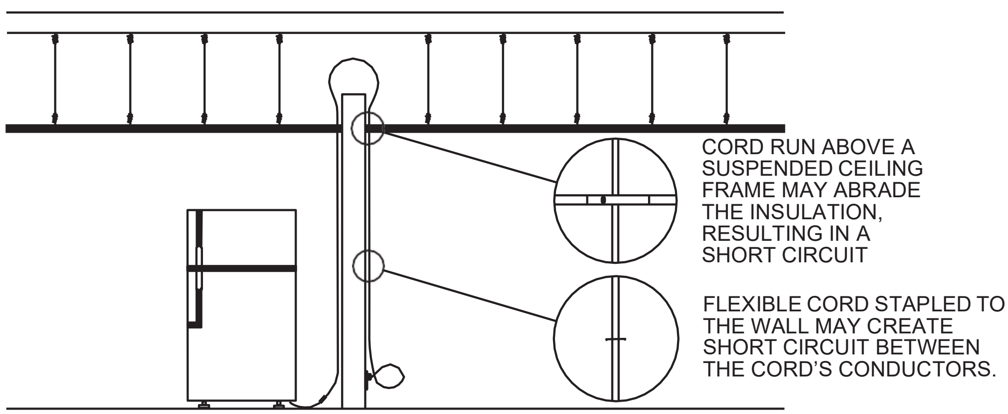
Overloading of extension cords causes an increase in the conductor's temperature. This increase in temperature can exceed the temperature rating of the conductor insulation, causing it to melt, decompose or burn. The burning insulation can ignite other combustible materials. The resulting loss of conductor insulation can also cause a short circuit or ground fault that can act as a source of ignition. The buildup of heat in an extension cord is often made worse by excessive cord length and by the insulating effect of rugs that often cover extension cords. Extension cords are much more susceptible to physical damage than permanent wiring. Damage to extension cords increases the likelihood of shorts and poor connections, both of which can cause a fire.

In addition to the fire hazard, extension cords pose a tripping hazard to occupants and, when damaged, can pose an electrical shock hazard. Securing flexible cords to a wall baseboard, door jambs, etc., with nails, staples or other fasteners to eliminate tripping hazards can create another dangerous condition by pinching or piercing the cord and causing shorts or faults that could lead to ignition.

Also, as a way of limiting the use of extension cords, their use is restricted to portable appliances. The reference to "portable" primarily denotes smaller, often-relocated appliances, such as a fan or a power tool. Extension cords must not be used with major appliances or equipment, such as refrigerators, which are obviously not portable. See the definitions for "Portable equipment" and "Portable appliance" in NFPA 70 Sections 520.2 and 550.2, respectively.

The last sentence addresses the fire and shock hazard associated with the use of indoor use extension cords in outdoor environments.

Commentary Figure 603.6—Examples of Unacceptable Practices



603.6.1 Application and use. Extension cords shall be plugged directly into an *approved* receptacle, relocatable power tap or current tap and, except for *approved* multiplug extension cords, shall serve only one portable appliance.

C This restriction means that multiple extension cords must not be connected to one another. Additionally, extension cords are limited to one appliance unless they are specifically approved multiplug extension cords.

603.6.2 Ampacity. The ampacity of the extension cords shall be not less than the rated ampacity of the portable appliance supplied by the cord.

C Although most building occupants may have difficulty understanding ampacity, which is the amount of electrical current that a particular conductor is capable of handling without exceeding its temperature limits, it is extremely important. If an appliance demands a higher electrical current than the extension cord is intended to handle, the extension cord will be overloaded, causing overheating and a potential fire. Familiarity with the types of extension cords available for sale and the general relationships to common appliances will help with the enforcement of this section.

603.6.3 Maintenance. Extension cords shall be maintained in good condition without splices, deterioration or damage.

C Where extension cords are damaged, they become potential shock hazards and a source of ignition.

603.6.4 Grounding. Extension cords shall contain an equipment grounding conductor where serving portable appliances required to be connected to an equipment grounding conductor.

C If an extension cord serves an appliance that requires a grounding conductor to avoid potential shock, that cord must provide a grounding conductor.

603.7 Electrical motors. Electrical motors shall be maintained free from excessive accumulations of oil, dirt, waste and debris.

C Internal heating is commonly associated with the operation of electrical motors. Excessive accumulations of dust, oil, grease, dirt or other debris may be easily ignited by the internal frictional heating of electrical motor components.

603.8 Temporary wiring. The use of temporary wiring for electrical power and lighting installations shall not exceed a period of 90 days. Temporary wiring methods shall meet the applicable provisions of NFPA 70.

Exception: Temporary wiring for electrical power and lighting installations complying with the applicable provisions of NFPA 70 is permitted during periods of construction, remodeling, repair or demolition of buildings, structures, equipment or similar activities.

C In some cases, because of a specific need and the temporary nature of the need, temporary wiring is allowed for a period of not more than 90 days. This allowance is primarily aimed at needs such as for holiday lighting. The exception allows temporary wiring to exceed 90 days for certain activities, such as remodeling or general construction of a building. Section 590 of NFPA 70 contains specific requirements for temporary wiring.

The term “temporary wiring” is not referring here to the use of power taps or extension cords (see commentary, Sections 603.5 and 603.6). The requirements for temporary wiring are less restrictive than those for permanent wiring but are much more rigorous than the requirements for the use of RPTs and extension cords.

603.8.1 Attachment to structures. Temporary wiring attached to a structure shall be protected from physical damage and supported on insulators spaced not more than 10 feet (3048 mm) apart.

C Where wiring is specifically attached to a structure, care must be taken to make sure that the attachment will not damage the wiring in a way that would cause resistance heating in localized areas of the wiring (see commentary, Section 603.6).

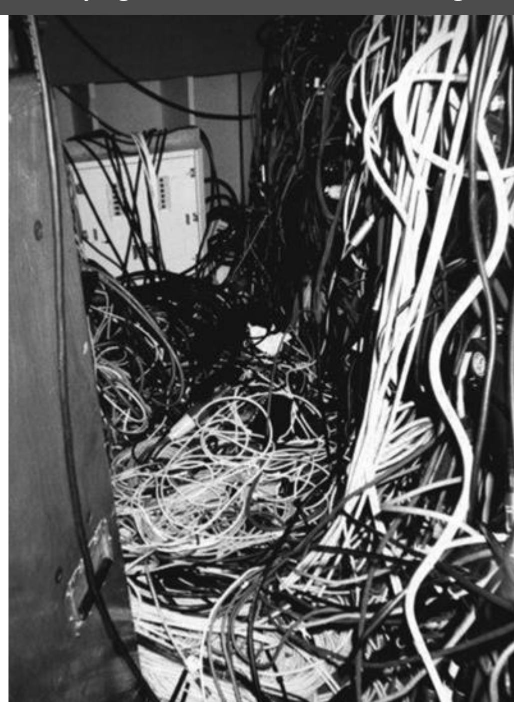
603.9 Abandoned wiring in plenums. Abandoned cables in plenums that are able to be accessed without causing damage, or requiring demolition to the building, shall be tagged for future use or shall be removed.

C This section introduces a concept that has long been in NFPA 70 and NFPA 90A—that plenums (see Chapter 2 commentary to the definition of “Plenum”) are intended specifically to be a part of the air distribution system. Plenums are also used legitimately for stringing communications and data cables as well as utility pipes, sprinkler pipes and similar items. However, such items often are not removed from the plenum when they become obsolete. For example, when an updated data system is installed in a facility (which can occur every 18 to 24 months), it is not unusual for old wires to be cut out of the system but left in place with a new wiring system added on top of them (see Commentary Figure 603.9).

The suspended ceiling tile systems that often enclose plenums are not intended to support any significant weight and can, therefore, easily be overwhelmed by the added weight of storage or abandoned materials such as cables (see commentary, IBC Section 808.1.1.1). Recently, a plenum space fuel load study showed how the safety of firefighters is compromised by the weight of these abandoned cables. It pointed out that plenum space fuel loads and wiring issues are serious concerns for firefighters during interior firefighting operations and its key recommendation was that abandoned wiring be removed.

Although the primary reason to require the removal of abandoned materials in plenums is the elimination of unnecessary weight, fire safety considerations should also be taken into account, as should the practical consideration that the HVAC design airflow through the plenum could be adversely affected. Such a requirement was long believed not to be enforceable primarily because fire code inspectors would rarely spend their time looking into plenums in existing buildings. However, the danger presented by the fire load of storage or abandoned materials is well-documented. Thus, there should be no significant difficulty in having inspectors identify the existence of abandoned products—especially abandoned cables—classify them as storage and require their removal. Note that only abandoned cable that can be “accessed without causing damage or requiring demolition” must be removed because there is no intent to cause potential damage to the building or facility by attempting to remove cables or circuits that are strung through walls, floors or other building elements. Where some cables are left in place because they are intended to be reused at a future date, they must be clearly identified as such by affixing a tag. See also Section 315.6.

Commentary Figure 603.9—Abandoned Wiring in Plenum



SECTION 604—ELEVATOR OPERATION, MAINTENANCE AND FIRE SERVICE KEYS

604.1 General. Where elevators and conveying systems are installed, they shall comply with Chapter 30 of the *International Building Code* and Sections 604.2 through 604.7 of this code.

C This section simply correlates the elevator requirements with IBC Chapter 30 and clarifies applicability of Section 604. Section 604 addresses certain aspects of both new and existing elevators.

604.2 Emergency operation. Existing elevators with a travel distance of 25 feet (7620 mm) or more shall comply with the requirements in Chapter 11. New elevators shall be provided with Phase I emergency recall operation and Phase II emergency in-car operation in accordance with ASME A17.1/CSA B44.

C Elevators are often used as a tool by emergency personnel when responding to fires and other emergencies. Due to these needs, the elevators must be capable of providing certain functions such as recall and emergency operation.

This section establishes requirements for both new and existing elevators. Existing elevators that travel 25 feet (635 mm) or more above or below the main level must, as a minimum, be equipped with emergency operation capabilities that comply with ASME A17.3 as required by Section 1103.3.2. New elevator installations are held to more restrictive guidelines for increased cost effectiveness and must have both emergency recall (Phase I) and emergency in-car operation (Phase II) to comply with ASME A17.1 for any amount of travel distance. The ASME standards are safety codes for elevators and escalators: ASME A17.3 is for existing elevators and ASME A17.1 for new elevator installations.

604.3 Standby power. In buildings and structures where standby power is required or furnished to operate an elevator, standby power shall be provided in accordance with Section 1203. Operation of the system shall be in accordance with Sections 604.3.1 through 604.3.4.

C This section states how standby power is to be supplied to elevators when required by other sections, such as Sections 1009.4.1 and 1009.5. The requirements from this section are the same as those located in IBC Chapter 30. Section 1203 addresses the overall requirements for emergency and standby power, including the relevant standards and other performance criteria.

604.3.1 Manual transfer. Standby power shall be manually transferable to all elevators in each bank.

C This section requires that all elevators in each bank of elevators must be equipped for manual transfer to standby power. All elevators, however, would not need to operate on standby power at the same time; they simply need to have the capability to transfer the power manually.

604.3.2 One elevator. Where only one elevator is installed, the elevator shall automatically transfer to standby power within 60 seconds after failure of normal power.

C Where a building has a single elevator, it must be automatically transferred to standby power within 60 seconds. The 60 seconds is a reflection of the NFPA 70 requirements for standby power.

604.3.3 Two or more elevators. Where two or more elevators are controlled by a common operating system, all elevators shall automatically transfer to standby power within 60 seconds after failure of normal power where the standby power source is of sufficient capacity to operate all elevators at the same time. Where the standby power source is not of sufficient capacity to operate all elevators at the same time, all elevators shall transfer to standby power in sequence, return to the designated landing and disconnect from the standby power source. After all elevators have been returned to the designated level, not less than one elevator shall remain operable from the standby power source.

C Where there is more than one elevator operating off a common system, the elevators must all be transferred to standby power within 60 seconds. Where only one elevator needs to be available during a loss of power, all elevators must still have the ability to run on standby power. Specifically, all elevators must initially connect to the standby power system and then, in sequence, return to the designated floor where all but one would be disconnected from the standby power.

604.3.4 Machine room ventilation. Where standby power is connected to elevators, the machine room ventilation or air conditioning shall be connected to the standby power source.

C This section reduces the likelihood that the equipment running the elevators will overheat during a loss of power because standby power is also required to power the ventilation or air conditioning for those areas. Note that this would also address machine spaces, control rooms and control spaces.

[BE] 604.4 Emergency signs. An *approved* pictorial sign of a standardized design shall be posted adjacent to each elevator call station on all floors instructing occupants to use the exit *stairways* and not to use the elevators in case of fire. Where elevators are not a component of the *accessible means of egress*, the sign shall read: "IN CASE OF FIRE, ELEVATORS ARE OUT OF SERVICE. USE EXIT."

Exception: The emergency sign shall not be required for elevators that are used for occupant self-evacuation in accordance with Section 3008 of the *International Building Code*.

C Because of the needs of firefighters and the possible risks posed by the use of elevators during a fire, signage is required that prohibits the use of elevators by building occupants during a fire emergency. The need to evacuate all occupants regardless of their physical abilities is becoming a more important issue and, in some cases, elevators are specifically used for such purposes. Section 1009.4 requires standby power for the purpose of providing the means of accessible egress. It is important to note that it is not intended that these elevators be used for self-evacuation but instead are for assisted rescue by emergency responders.

604.5 Maintenance of elevators. Elevator features and lobbies required by Section 3006 of the *International Building Code* shall be inspected, tested and maintained in accordance with Sections 604.5.1 through 604.5.4.

C This section introduces the maintenance-related requirements associated with elevators.

604.5.1 Fire service access elevators and lobbies. Where fire service access elevators are required by Section 3007 of the *International Building Code*, fire service access elevator fire protection and safety features shall be maintained and lobbies required by Section 3007 of the *International Building Code* shall be maintained free of storage and furniture.

C The fire service access elevator in high-rise buildings over 120 feet (3657 mm) in height above fire department vehicle access, as required by IBC Section 403.6.1, is a tool which enhances firefighters' abilities to gain access to the fire floor and undertake necessary operational staging activities in a protected area. This section complements the provisions of the fire service access elevator provisions contained in IBC Section 3007 by requiring that fire service access elevator lobbies always be fully and immediately accessible by the fire service without impediment by storage or furniture and the fire load that some items might present. It is not unusual, especially in Group R-1 occupancies, to see tables, chairs, sofas or the like placed in areas opposite elevator entrances for aesthetic purposes in what may now be a required fire service access elevator lobby. This section would prohibit that practice. It should also be noted that the provisions of this section apply to noncombustible storage and furnishings, as well as combustible (see commentaries, IBC Sections 403.6.1 and 3007).

This section also requires that the fire protection and safety features associated with such elevators are appropriately maintained to ensure functionality during an emergency.

604.5.2 Occupant evacuation elevators and lobbies. Where occupant evacuation elevators are provided in accordance with Section 3008 of the *International Building Code*, occupant evacuation elevator fire protection and safety features shall be maintained and lobbies required by Section 3008 of the *International Building Code* shall be maintained free of storage and furniture.

C This section is similar in intent to Section 604.5.1 for fire service access elevators except that the focus is providing adequate space for those awaiting evacuation via the elevators. If storage or furnishings are allowed, it takes away from that space and also will cause more people to wait outside the protected lobby. Placing combustibles within the lobby also creates potential fire hazards.

This section also requires that the fire protection and safety features associated with such elevators are appropriately maintained to ensure functionality during an emergency.

604.5.3 Storage within elevator lobbies. Where hoistway opening protection is required by Section 3006.2 of the *International Building Code*, elevator lobbies shall be maintained free of storage.

C This section is intended to prohibit storage where an elevator lobby is provided to comply with the provisions for hoistway opening protection. Where hoistway opening protection is required, allowing storage within a lobby would add fuel load to a protected space, which is counter to the intent of the requirements.

604.5.4 Water protection of hoistway enclosures. Methods to prevent water from infiltrating into a hoistway enclosure required by Sections 3007.3 and 3008.3 of the *International Building Code* shall be maintained.

C IBC Sections 3007.3 and 3008.3 require that hoistway openings be protected from water originating from sprinklers activated outside of the elevator lobby. This is critical to extend the use of elevators during a fire. The methods used are not prescribed in the IBC but often will require some level of maintenance to make sure this feature will be operational and available during a fire.

604.6 Elevator keys. All elevators shall be provided with elevator car door and firefighter service keys in accordance with Sections 604.6.1 through 604.6.2.4.

C This section clarifies which requirements apply to elevator keys.

604.6.1 Elevator key location. Keys for the elevator car doors and firefighter service keys shall be kept in an *approved* location for immediate use by the fire department.

C The fire service often responds to elevator emergencies and other emergencies that require the use or operation of the elevators and associated components. Most elevators will have at least five or six key-switched functions, including those for standby power transfer, Phase I emergency elevator recall (hall switch located in the elevator lobby at the designated level), Phase II in-cab emergency operation, inspection function, normal lighting and fan operation. Ready access to elevator keys is an important factor in the timely and efficient response to emergency situations. Because experience has shown that this important safety element is often overlooked, this section requires that elevator keys be kept in a location approved by the fire code official to ensure immediate access. Many elevator installers will provide, as part of their package, an elevator key box (see Commentary Figure 604.6.1), which will accommodate all of the required operating keys, plus an appropriate hoistway access key or tool for fire department use in accessing the hoistway in case of an elevator emergency. Elevator inspection reports should note the presence and locations of the keys. Inspectors may verify the availability of the keys during periodic fire safety inspections and prefire planning surveys.

Commentary Figure 604.6.1—Typical Elevator Key Box

604.6.2 Standardized fire service elevator keys. Buildings with elevators equipped with Phase I emergency recall, Phase II emergency in-car operation, or a fire service access elevator shall be equipped to operate with a standardized fire service elevator key *approved* by the *fire code official* or a standardized key in accordance with ASME A17.1/CSA B44.

Exception: The *owner* shall be permitted to place the building's nonstandardized fire service elevator keys in a key box installed in accordance with Section 506.1.2.

C When fire departments and other emergency response agencies respond to emergencies, their ability to quickly access the location of the emergency can be the deciding factor of a successful response. Elevators are increasingly being relied on for emergency operations, and their importance has been highlighted by adding Section 3007 to the IBC, requiring the installation of fire service access elevators and providing requirements for the installation of occupant evacuation elevators. One of the difficulties the fire service and other emergency response agencies have when accessing facilities and attempting to use elevators is the number of nonstandardized keys that may not be available at the time of response. Even when emergency responders are provided the necessary keys in case of response, the correct key may have to be identified from a large collection of keys for any one building. In larger jurisdictions, the sheer number of keys makes possession of them unwieldy for the emergency responders. An elevator key is an important tool to firefighters. The key allows firefighters to access the interior of the shaft housing an elevator cabin to initiate rescue in case of a malfunction or the loss of building power. More often, an elevator key is used to capture and control an elevator in emergencies. This section establishes requirements for elevator keys used by the fire service that will apply to only those buildings that have elevators with Phase I or Phase II emergency service or to those buildings equipped with a fire service access elevator. Note that this section is aimed at providing consistency within a jurisdiction to facilitate successful emergency responses.

The exception recognizes that many jurisdictions have elevators built prior to the introduction of a standard elevator key and as a result, each building with an elevator requires its own elevator key. The exception authorizes the jurisdiction to allow the installation of a key box to house nonstandardized elevator keys (see commentary, Section 506.1.2).

Note that this section recognizes that such standardized keys can either be approved by the jurisdiction or in accordance with ASME A17.1/CSA B44. The reference to the standard provides more flexibility and needed consistency between the IFC and ASME A17.1/CSA B44.

604.6.2.1 Requirements for standardized fire service elevator keys. Standardized fire service elevator keys shall comply with all of the following:

1. All fire service elevator keys within the jurisdiction shall be uniform and *approved* in accordance with Section 604.6.2. Keys shall be cut to a uniform key code.
2. Fire service elevator keys shall be of a patent-protected design to prevent unauthorized duplication.
3. Fire service elevator keys shall be factory restricted by the manufacturer to prevent the unauthorized distribution of key blanks. Uncut key blanks shall not be permitted to leave the factory.
4. Fire service elevator keys subject to these rules shall be engraved with the words "DO NOT DUPLICATE."

C This section requires that keys for any new elevators installed in the jurisdiction be approved in accordance with Section 604.6.2, regardless of elevator manufacturer or model. Standardizing the type of elevator key creates a consistent arrangement for firefighters who utilize the elevators for EMS incidents or for the deployment of personnel and equipment in a multiple-story firefighting operation. The keys must be manufactured to prevent unauthorized duplication. Consistency between jurisdictions may be important where there are mutual aid agreements.

604.6.2.2 Access to standardized fire service keys. Access to standardized fire service elevator keys shall be restricted to the following:

1. Elevator owners or their authorized agents.
2. Elevator contractors.
3. Elevator inspectors of the jurisdiction.
4. *Fire code officials* of the jurisdiction.
5. The fire department and other emergency response agencies designated by the *fire code official*.

C Access to or possession of a key that can take control of an elevator in a building is an area of vulnerability for buildings that was not previously addressed with simple key designs being utilized. In addition to the limitations placed on the unauthorized duplication and distribution of standardized elevator keys in Section 604.6.2.3, this section specifies exactly to whom the keys will be accessible. This list includes only those persons who have a vested interest in the building, the elevator equipment or the protection of the building.

604.6.2.3 Duplication or distribution of keys. A person shall not duplicate a standardized fire service elevator key or issue, give, or sell a duplicated key unless in accordance with this code.

C This section provides for a level of security for the standardized key by prohibiting its unauthorized duplication and distribution. Note that Section 604.6.2.1 requires that keys be stamped or embossed with a duplication prohibition warning.

604.6.2.4 Responsibility to provide keys. The building *owner* shall provide up to three standardized fire service elevator keys where required by the *fire code official*, upon installation of a standardized fire service key switch or switches in the building.

C This section establishes the responsibility of the building owner to provide at least three keys where an elevator is equipped with Phase I and II firefighter service or is a fire service access elevator. Multiple keys allow fire companies to use multiple elevators to carry personnel and equipment to the fire floor or swiftly remove multiple injured or otherwise incapacitated persons from upper or lower floors.

604.7 Storage. Storage is prohibited in elevator cars or elevator machine rooms.

Exceptions:

1. Blankets used for protection of elevator cab walls during construction or renovation.
2. Materials necessary for the operation and maintenance of the elevator equipment.

C The purpose of this section is to clarify that elevator cars and machine rooms are not to be used for storage. Exception 1 addresses blankets that are used for protecting the elevator cab walls when activities such as building construction and renovations are occurring, which is a common practice. Exception 2 is aimed more at the machine room and allows materials associated with the maintenance of the elevator machinery.

SECTION 605—FUEL-FIRED APPLIANCES

605.1 General. The design, construction, installation, operation, alteration, repair and maintenance of nonportable gas-fired appliances and systems shall comply with the provisions of this section and the *International Fuel Gas Code*. The design, construction, installation, operation, alteration, repair and maintenance of nonportable solid fuel-fired and oil-fired appliances and systems shall comply with the provisions of this section and the *International Mechanical Code*.

C This section sets the scope as to what type of fuel-fired appliances and systems are regulated and by which code. More specifically, there is a distinction made between nonportable appliances and systems and portable appliances. This section applies to all types of fuel-fired appliances and systems, as applicable, but where the appliance or system is nonportable, either the IFGC or IMC will also apply. The IFGC and IMC do not cover portable appliances.

605.1.1 Installation of nonportable fuel-fired appliances. The installation of nonportable fuel-fired appliances shall be made in accordance with the manufacturer's installation instructions and applicable federal, state and local rules and regulations.

C Compliance with the appliance manufacturer's installation instructions is a fundamental requirement of all I-Codes, and those instructions are an enforceable extension of the code. Federal, state, county or municipal laws might supersede part of the installation instructions or could be applied in addition to the requirements in the instructions.

605.1.2 Electrical wiring and equipment. Electrical wiring and equipment used in connection with fuel-fired appliances and equipment shall be installed and maintained in accordance with Section 603 and NFPA 70.

C Section 603 contains provisions intended to mitigate fire hazards and shock hazards associated with the use of existing appliances. NFPA 70 covers the installation of electrical appliances.

605.1.3 Fuel oil. The grade of fuel oil used in an oil burner shall be that for which the oil burner is *approved* and as stipulated by the oil burner manufacturer's instructions. Oil containing gasoline shall not be used. Waste crankcase oil shall be an acceptable fuel in Group F, M and S occupancies where utilized in equipment *listed* and *labeled* for use with waste oil and where such equipment is installed in accordance with the manufacturer's instructions and the terms of its listing.

C Different grades of fuel oil have different viscosities and chemical makeup. A burner and fuel mismatch could result in poor combustion, sooting and burner component failure. Oil burners are not designed to burn oil contaminated with chemicals of higher volatility. Appliances that consume used engine oil are allowed only in occupancies of low-occupant density (those without sleeping rooms) and where such appliances will likely be monitored and maintained by facility personnel. Used engine oil appliances use a specialized type of atomizing oil burner designed to burn dirty waste oil collected from internal combustion engine maintenance operations.

605.1.4 Access. The installation of fuel-fired equipment shall be provided with access to equipment for cleaning hot surfaces; removing burners; replacing motors, controls, air filters, chimney and vent connectors, draft regulators and other working parts; and for adjusting, cleaning and lubricating parts.

C The IFGC and the IMC require access for the initial installation as well as for the life of the appliance. Safe operation depends on observation and maintenance, which depend on adequate access to the appliances. In order for the installation to be considered as providing appropriate access, personnel should be able to reach it without having to remove building elements or obstacles of any kind or use climbing aids.

605.1.5 Testing, diagrams and instructions. After installation of the fuel-fired equipment, operation and combustion performance tests shall be conducted to determine that the equipment is in proper operating condition and that all accessory equipment, controls, and safety devices function properly.

C Testing of an appliance after installation is also required by the IMC and the appliance manufacturer's installation instructions.

605.1.5.1 Diagrams. Contractors installing industrial oil-burning systems shall furnish not less than two copies of diagrams showing the main oil lines and controlling valves, one copy of which shall be posted at the oil-burning equipment and another at an *approved* location that will be available in case of emergency.

C For large systems, the piping and control valve layout may be complicated and extensive. In the event of an emergency, facility personnel or firefighters might need to be able to access control valves to protect piping and to limit the fire hazard.

605.1.5.2 Operating instructions. After completing the installation, the installer shall instruct the *owner* or operator in the proper operation of the equipment. The installer shall furnish the *owner* or operator with the manufacturer's operating instructions.

C Appliances are more likely to be properly (safely) operated and maintained if the owner or operator is instructed in the operation of the equipment.

605.1.6 Clearances. Working clearances between fuel-fired appliances and electrical panelboards and equipment shall be in accordance with NFPA 70. Clearances between oil-fired equipment and oil supply tanks shall be in accordance with NFPA 31.

C NFPA 70 requires working clearances around electrical equipment for protection of personnel. NFPA 31 requires clearances between appliances and equipment and oil-supply tanks to protect the oil tank from excessive heat and to lessen the fire hazard from any oil leakage.

605.2 Abatement of unsafe conditions. The *fire code official* is authorized to order that measures be taken to prevent the operation of any existing stove, oven, furnace, incinerator, boiler or any other heat-producing device or appliance found to be defective or in violation of code requirements for existing appliances after giving notice to this effect to any person, *owner*, firm or agent or operator in charge of the same. The *fire code official* is authorized to take measures to prevent the operation of any device or appliance without notice when inspection shows the existence of an immediate fire hazard or when imperiling human life. The defective device shall remain withdrawn from service until all necessary repairs or alterations have been made or replaced in accordance with Section 605.1.

C When a heat-producing appliance or system is determined to be unsafe, the fire code official is required to notify the owner or agent of the building as the first step in correcting the difficulty. This notice may describe the repairs and improvements necessary to correct the deficiency and keep the system in operation or require the unsafe equipment or system to be removed or replaced. The notice must specify a time frame in which the corrective actions must occur. Additionally, the notice should require the immediate response of the owner or agent.

If the owner or agent is not available, public notice of the declaration would be enough to comply with this section. The fire code official may also determine that the system must be disconnected to correct an unsafe condition and must give written notice to that effect; however, an immediate disconnection can be ordered if it is essential for protection of public health and safety.

605.2.1 Chimneys and appliances. Chimneys, vents, incinerators, smokestacks or similar devices for conveying smoke or hot gases to the outer air and the appliances such as stoves, furnaces, fireboxes or boilers to which such devices are connected, shall be maintained so as not to create a fire hazard.

C A primary function of the code is to reduce or eliminate fire hazards through proper maintenance of appliances and systems that are potential fire and life safety hazards.

605.2.1.1 Masonry chimneys. Masonry chimneys that, upon inspection, are found to be without a flue liner and that have open mortar joints which will permit smoke or gases to be discharged into the building, or which are cracked as to be dangerous, shall be repaired or relined with a *listed* chimney liner system installed in accordance with the manufacturer's instructions and the *International Mechanical Code* or a flue lining system installed in accordance with the requirements of the *International Building Code* and appropriate for the intended class of chimney service.

C Based on a finding during inspection, a new flue liner may be required. This section is ensuring that the flue liner meets the appropriate code and listing requirements, as applicable. See Section 2113 of the IBC, Section 801.16 of the IMC, and Sections 501.12, 501.13 and 503.5 of the IFGC for information on masonry chimney liners and masonry chimneys.

605.2.1.2 Metal chimneys. Metal chimneys that are corroded or improperly supported shall be repaired or replaced in accordance with the *International Mechanical Code*.

C See the commentary to Section 605.3.

605.2.1.3 Decorative shrouds. Decorative shrouds installed at the termination of factory-built chimneys or vents shall be removed except where such shrouds are *listed* and *labeled* for use with the specific factory-built chimney system and are installed in accordance with the chimney or vent manufacturer's instructions and the *International Mechanical Code* or *International Fuel Gas Code*.

C This section is retroactive in that it requires removal of a previously installed trim item. IMC Section 805.7 and IFGC Section 503.5.4 prohibit the installation of decorative shrouds not meeting the listing criteria. The code addresses those noncomplying shrouds that were illegally installed (see commentary, IMC Section 805.7 and IFGC Section 503.5.4).

605.2.1.4 Factory-built chimney and vent systems. Existing factory-built chimneys and vent systems that are damaged, corroded or improperly supported shall be repaired or replaced in accordance with the *International Mechanical Code*.

C Defective or inadequately supported chimneys could leak flue gas and could fail structurally, resulting in separation, collapse, a fire hazard and a life safety hazard. This section is consistent with the maintenance focus of the code. Such repairs and replacement need to comply with the IFGC or IMC, as applicable.

605.2.1.5 Connectors. Existing chimney and vent connectors that are damaged, corroded or improperly supported shall be repaired or replaced in accordance with the *International Mechanical Code*.

C See the commentary to Section 605.2.1.4.

605.3 Chimneys and vents. Masonry chimneys shall be constructed in accordance with the *International Building Code*. Factory-built chimneys and vent systems serving solid-fuel-fired appliances or oil-fired appliances shall be installed in accordance with the *International Mechanical Code*. Metal chimneys shall be constructed and installed in accordance with the *International Mechanical Code*. Factory-built chimneys and vent systems serving gas-fired appliances shall be installed in accordance with the *International Fuel Gas Code*.

C The IBC regulates masonry chimney construction in Chapter 21. Factory-built chimneys and vent systems are regulated by IMC Section 805. Metal chimneys are distinct from factory-built chimneys, are industrial occupancy related (e.g., smokestacks) and are regulated by NFPA 211 (see commentary, IMC Section 806.1). Chimneys and vent systems associated with gas-fired appliances are required to comply with IFGC Section 506, which references UL 103 for flue gases having a temperature not greater than 1,000°F (538°C) and UL 959 for those having flue gas temperatures greater than 1,000°F (538°C).

605.4 Fuel oil storage systems. Fuel oil storage systems shall be installed and maintained in accordance with this code. Tanks and fuel oil piping systems shall be installed in accordance with Chapter 13 of the *International Mechanical Code*.

C The scope of this section is focused on fuel oil storage systems, not on generators or fire pumps. However, Section 605.4.2.3 allows compliance with Section 605.4.2.2 for fuel oil that supplies fuel oil burning equipment, generators or fire pumps installed in accordance with Section 605.4.2.5. The IMC regulates installation of the fuel oil distribution piping system.

605.4.1 Fuel oil storage in outside, above-ground tanks. Where connected to a fuel oil piping system, the maximum amount of fuel oil storage allowed outside above ground without additional protection shall be 660 gallons (2498 L). The storage of fuel oil above ground in quantities exceeding 660 gallons (2498 L) shall comply with NFPA 31.

C To limit the potential fire hazard resulting from oil spillage, the code sets a quantity limitation on unprotected storage. NFPA 31 requires protection, such as spill containment, for storage in excess of 660 gallons (2498 L). The storage of 660 gallons (2498 L) is allowed to be in any configuration of containers that does not exceed a total of 660 gallons (2498 L).

605.4.1.1 Approval. Outdoor fuel oil storage tanks shall be in accordance with UL 80, UL 142, UL 142A or UL 2085.

C This section establishes the applicable standard for outdoor fuel oil storage tanks. Section 605.4.2.1 addresses indoor tanks and has one additional allowable standard, UL 80, which is not appropriate for outdoor applications.

605.4.2 Fuel oil storage inside buildings. Fuel oil storage inside buildings shall comply with Sections 605.4.2.2 through 605.4.2.8 or Chapter 57.

C This section introduces Sections 605.4.2.1 through 605.4.2.8, which contain requirements for controlling the hazards associated with the storage of fuel oil inside of buildings. Fuel oil is defined in the IMC as "Kerosene or any hydrocarbon oil having a flash point not less than 100°F (38°C)." This would include Number 2 diesel fuel, which is classified as either a Class II or IIIA combustible liquid, depending on the refiner and the region where the fuel will be used. Number 2 diesel fuel is commonly used as a fuel source for diesel-driven electric generators.

The indicated sections have been revised to allow larger amounts of fuel oil in storage inside of buildings in response to increased fuel oil storage requirements for switch and data centers and similar telecommunications facilities. Such facilities are constructed to house large numbers of computers to serve as remote data centers for the protection of computer data or data switches for internet providers. To increase the likelihood that the data is always available, design professionals place a great deal of emphasis on the building's electrical power supply. These types of uses consume large amounts of electrical power. Accordingly, large generators are installed to ensure that service is not disrupted. These generators have fairly demanding fuel requirements. Consider, for example, that a single 2-megawatt (2 million watt) generator can have a fuel consumption rate of 3-4 gallons/minute under full electrical load conditions. As a result, these facilities require a large volume of fuel. The fuel storage is commonly located inside of a building because many of these facilities are located in commercial or other densely populated areas of a community.

605.4.2.1 Approval. Indoor fuel oil storage tanks shall be in accordance with UL 80, UL 142, UL 142A or UL 2085.

C This section establishes the applicable standard for indoor fuel oil storage tanks. Section 605.4.1.1 addresses outdoor tanks.

605.4.2.2 Quantity limits. One or more fuel oil storage tanks containing Class II or III *combustible liquid* shall be permitted in a building. The aggregate capacity of all tanks shall not exceed the following:

1. 660 gallons (2498 L) in unsprinklered buildings, where stored in a tank complying with UL 80, UL 142, UL 142A or UL 2085.
2. 1,320 gallons (4996 L) in buildings equipped with an *automatic sprinkler* system in accordance with Section 903.3.1.1, where stored in a tank complying with UL 142 or UL 142A. The tank shall be *listed* as a secondary containment tank, and the secondary containment shall be monitored visually or automatically.
3. 3,000 gallons (11 356 L) in buildings equipped with an automatic sprinkler system in accordance with Section 903.3.1.1, where stored in protected above-ground tanks complying with UL 2085 and Section 5704.2.9.7. The tank shall be *listed* as a secondary containment tank, as required by UL 2085, and the secondary containment shall be monitored visually or automatically.

C This section correlates with Table 5003.1.1(5) and Section 5701.2, Item 4 of this code and IBC Table 307.1.1. This section essentially provides flexibility to allow more combustible fuel oil in buildings beyond that allowed by maximum allowable quantities (MAQs) for other code applications. These amounts are essentially exceptions to the maximum allowance quantities of 120 gallons for Class II liquids and 330 Gallons for Class IIIA liquids in Table 5003.1.1(1) and IBC Table 307.1(1). Note that these amounts would be allowed to be doubled where the building is provided with a sprinkler system throughout. In order to apply this section, such fuel supplies are required to be connected to a closed fuel oil piping system. This would apply to most oil-fired stationary equipment in industrial, commercial and residential occupancies. Note that this provision applies only to the aggregate storage of fuel oil and does not include the capacity of the piping system.

Where the need for an aggregate indoor storage quantity of fuel oil connected to a closed fuel oil piping system exceeds the MAQs in Table 5003.1.1(1) and IBC Table 307.1(1), the allowances of this section may be applied. There are three limits based on whether sprinklers are installed and the type of tank used as follows:

1. 660-gallon limit. Item 1 is applicable to buildings that are not equipped with a sprinkler system throughout. Such buildings are allowed 660 gallons, which relates to previous editions. However, in the past, the limit was to both buildings, with or without sprinkler systems. Tank listing allowed includes all three permitted in Section 605.4.2.1.
2. 1,320-gallon limit. Item 2 provides credit to fuel storage amounts in buildings equipped throughout with sprinklers and would permit 1,320 gallons. Essentially, this is a 100 percent increase for an automatic sprinkler system, similar to what Table 5003.1.1(1) and IBC Table 307.1(1) provide. This allowance also requires use of a UL 142-listed tank with secondary containment and a method to monitor the secondary containment.
3. 3,000-gallon limit. Item 3 addresses the allowance for using protected above-ground tanks where automatic sprinklers are also provided. Sprinklered buildings or rooms containing such tanks are allowed increases based on the safety feature being provided. Item 3 allows 3,000 gallons where a UL 2085-protected, above-ground tank is used and sprinklers are provided throughout the building. These tanks have extensive regulations in Chapter 57, and the

listing requirements further document their safety. Included in the design requirements for these tanks are the ability to survive a 2-hour pool fire test conducted in accordance with the UL 1709 fire exposure protocol; a limitation that all penetrations must be made through the top of the tank (to avoid the risk of a gravity-fed leak that might be associated with a connection below liquid level); and that piping connected to the tank must be provided with anti-siphon controls where needed to prevent a siphon risk, among others. Similar to Item 2, a method of monitoring the secondary containment is required. See the commentary to the Section 202 definition of "Tank, protected above ground" and Section 5704.2.9.7.

605.4.2.3 Restricted use and connection. Tanks installed in accordance with Section 605.4.2 shall be used only to supply fuel oil to fuel-burning equipment, generators or fire pumps installed in accordance with Section 605.4.2.5. Connections between tanks and equipment supplied by such tanks shall be made using closed piping systems in accordance with the *International Mechanical Code*.

C This section makes it clear that the fuel oil storage quantity limits in Section 605.4.2.2 are applicable only to fuel oil supplies for oil-burning equipment, generators and fire pumps and then only when connected to a fuel oil piping system supplying such equipment. Fuel oil piping systems are regulated by IMC Chapter 13.

605.4.2.4 Applicability of maximum allowable quantity and control area requirements. The quantity of *combustible liquid* stored in tanks complying with Section 605.4.2 shall not be counted towards the maximum allowable quantity set forth in Table 5003.1.1(1), and such tanks shall not be required to be located in a *control area*.

C This section clearly states that the fuel oil quantities in Section 605.4.2, sometimes referred to as a "special quantity," are outside the scope of the MAQ per control area provisions of Chapter 50, including any requirement to install the tank in a control area in accordance with Section 5003.8.3. Table 5003.1.1(5) also correlates with these provisions and indicates that the MAQ provisions of the table do not apply to fuel oil storage complying with Section 605.4.2.

The "special quantity" concept originated in an ad-hoc hazardous materials committee constituted by one of the legacy model code groups. As the committee developed updated regulations pertaining to flammable and combustible liquids, it concluded that a closed fuel oil storage and piping system feeding oil-fired building service equipment should not be penalized for providing sufficient fuel oil on site for long-term operation of such systems. Accordingly, the ad-hoc committee, using NFPA 31 as a guide, created the original provisions of Section 605.4 and its subsections to provide that relief.

605.4.2.5 Installation. Tanks and piping systems shall be installed in accordance with Section 915 and Chapter 13, both of the *International Mechanical Code*, as applicable.

C IMC Section 915 addresses liquid-fueled internal combustion engines, turbines and permanently installed equipment and appliances powered by internal combustion engines and turbines. Engine-driven electrical generators for private use are becoming more popular, as are engine-driven cooling appliances and heat pumps. Such equipment is also used to power fire pumps, generators, water pumps, refrigeration machines and other stationary equipment. IMC Section 915 also references NFPA 37, which addresses the fire safety of this kind of equipment including requirements for enclosures, controls, fuel supplies, exhaust systems, cooling systems and combustion air.

IMC Chapter 13 complements this code's fuel oil storage provisions and regulates the design and installation of fuel oil piping systems. The regulations reference construction standards for above-ground and underground storage tanks, material standards for piping systems (both above ground and underground) and extensive requirements for the proper assembly of system piping and components.

605.4.2.6 Separation. Rooms containing fuel oil tanks for internal combustion engines shall be separated from the remainder of the building by *fire barriers, horizontal assemblies*, or both, with a minimum 1-hour *fire-resistance rating* with 1-hour fire-protection-rated *opening protectives* constructed in accordance with the *International Building Code*.

Exception: Rooms containing protected above-ground tanks complying with Section 5704.2.9.7 shall not be required to be separated from surrounding areas.

C This section addresses the separation of the tank from the remainder of the building. The requirement for 1-hour separation is not a new requirement. The requirement for separation is located in Section 4.1.2.1.1 and Section 6.3.5.1 of NFPA 37, both of which require 1-hour construction to separate internal combustion engines and associated fuel tanks up to 1,320 gallons from the remainder of the building in which they are located. Section 4.1.2.1.5 of NFPA 37 specifies that openings are protected with at least 1-hour opening protectives. This section eliminates the additional navigation to NFPA 37 through the reference to IMC Section 915 within Section 605.4.2.5.

The exception here, as in Section 605.4.2.2, recognizes the increased safety provided by protected above-ground tanks. See the commentary to Section 605.4.2.2 for a discussion of those tanks and their enhanced safety.

605.4.2.7 Spill containment. Tanks exceeding 60-gallon (227 L) capacity or an aggregate capacity of 1,000 gallons (3785 L) that are not provided with integral secondary containment shall be provided with spill containment sized to contain a release from the largest tank.

C Section 605.4.2 states that the installation must comply with Section 605 or Chapter 57. The designer has an option to select either design method. However, spill containment is found in Chapter 57, which the designer may or may not comply with. Therefore, this section specifies that spill containment is required where a single tank exceeds 60 gallons, or the aggregate exceeds 1,000 gallons. These thresholds are close to the thresholds in Section 5004.2.1 for spill containment of liquids.

The secondary containment is sized to contain the largest spill. Even for tanks located inside a sprinklered building, only the tank contents must be contained. This is consistent with the requirements found in Section 6.3.5.3 of NFPA 37. This section provides a more direct path to comply with NFPA 37.

605.4.2.8 Tanks in basements. Tanks in *basements* shall be located not more than two stories below *grade plane*.

C This section prohibits tank installations more than two stories below grade because the further an incident is below grade, the greater challenge for the fire department to mitigate it. This is consistent with the control area approach in Table 5003.8.3.2, which also limits control areas to two levels below grade.

605.4.3 Underground storage of fuel oil. The storage of fuel oil in underground storage tanks shall comply with UL 58 or UL 1316 and be installed in accordance with NFPA 31.

C Section 605.4 does not require that fuel oil tanks be installed underground; however, there may be circumstances under which such an installation is either desirable or advisable. This section sets out the appropriate listings for tanks and directs the code user to NFPA 31 for specific requirements applicable to the installation of underground combustible liquid storage tanks.

605.5 Heating appliances. Heating appliances shall be *listed* and shall comply with Sections 605.5.1 and 605.5.2.

C The IFGC and the IMC require that all space-heating appliances be listed and labeled.

605.5.1 Guard against contact. The heating element or combustion chamber shall be permanently guarded so as to prevent accidental contact by persons or material.

C The injury and ignition protection feature required by this section is typically designed into the appliance.

605.5.2 Heating appliance installation and maintenance. Heating appliances shall be installed and maintained in accordance with the manufacturer's instructions, the *International Building Code*, the *International Fuel Gas Code*, the *International Mechanical Code* and NFPA 70.

C Depending on the type of fuel utilized, appliance installation and maintenance can be subject to the requirements of multiple codes, including the IBC, the IFGC, the IMC and NFPA 70 in addition to the appliance manufacturer's instructions.

605.6 Unauthorized operation. It shall be a violation of this code for any person, user, firm or agent to continue the utilization of any device or appliance (the operation of which has been discontinued or ordered discontinued in accordance with Section 605.2) unless written authority to resume operation is given by the *fire code official*. Removing or breaking the means by which operation of the device is prevented shall be a violation of this code.

C Appliances or systems removed from service in accordance with Section 605.2 may be sealed or otherwise secured in a manner approved by the fire code official and may only be returned to service upon written authorization of the fire code official.

605.7 Incinerators. Commercial, industrial and residential-type incinerators and chimneys shall be constructed in accordance with the *International Building Code*, the *International Fuel Gas Code* and the *International Mechanical Code*.

C See the commentary to IMC Section 907.1, IFGC Section 606.1 and IBC Section 2113.

605.7.1 Residential incinerators. Residential incinerators shall be *listed* and *labeled* in accordance with UL 791.

C Residential incinerators have gone out of use today but may still be found in older homes and in rural areas. Where they are newly installed, they must be listed and labeled in accordance with UL 791.

605.7.2 Spark arrestor. Incinerators shall be equipped with an effective means for arresting sparks.

C Spark arrestor chimney caps are designed with a screened outlet that prevents the escape of burning embers and particles.

605.7.3 Restrictions. Where the *fire code official* determines that burning in incinerators located within 500 feet (152 m) of mountainous, brush or grass-covered areas will create an undue fire hazard because of atmospheric conditions, such burning shall be prohibited.

C The fire code official must determine whether incinerator use would present an unacceptable risk of wildfires in timber, brush and grass-covered areas. For wildland-urban interface areas, see the *International Wildland-Urban Interface Code*® (IWUIC®). The local air-quality agency may also have restrictions.

605.7.4 Time of burning. Burning shall take place only during *approved* hours.

C The jurisdiction must determine the periods that would be safe for burning and those that would be unsafe. Consideration must be given to daylight, prevailing winds, ambient temperatures, impact on air quality, moisture levels and presence of observers and supervisory personnel.

605.7.5 Discontinuance. The *fire code official* is authorized to require incinerator use to be discontinued immediately if the *fire code official* determines that smoke emissions are offensive to occupants of surrounding property or if the use of incinerators is determined by the *fire code official* to constitute a hazardous condition.

C The fire code official can prohibit incinerator use if it would be a nuisance or a health or fire hazard. Coordination with the local air-quality agency may also be necessary.

605.7.6 Flue-fed incinerators in Group I-2. In Group I-2 occupancies, the continued use of existing flue-fed incinerators is prohibited.

C This type of incinerator is hazardous and would not comply with current code requirements. This section is simply making sure that such incinerators are no longer used.

605.7.7 Incinerator inspections in Group I-2. Incinerators in Group I-2 occupancies shall be inspected not less than annually in accordance with the manufacturer's instructions. Inspection records shall be maintained on the premises and made available to the *fire code official* upon request.

C Incinerators pose a hazard to the building and its occupants if not properly maintained. This is more of a concern in Group I-2 occupancies where the residents are at particular risk from such hazards.

605.8 Gas meters. Above-ground gas meters, regulators and piping subject to damage shall be protected by a barrier complying with Section 312 or otherwise protected in an *approved* manner.

C Vehicle impact protection is necessary to prevent gas leakage resulting from impact damage to gas service equipment. Protection can be accomplished by location alone or by the construction of barriers as prescribed by Section 312. Barriers are required only where gas service equipment is located where vehicle impact is likely to occur.

SECTION 606—COMMERCIAL COOKING EQUIPMENT AND SYSTEMS

[M] 606.1 General. Commercial kitchen exhaust hoods shall comply with the requirements of the *International Mechanical Code*.

C Rather than including detailed commercial hood requirements here, the code simply references the IMC for hood design (see commentary, IMC Section 507).

[M] 606.2 Where required. A Type I hood shall be installed at or above all commercial cooking appliances and domestic cooking appliances used for commercial purposes that produce grease vapors.

Exceptions:

1. Factory-built commercial exhaust hoods that are *listed* and *labeled* in accordance with UL 710, and installed in accordance with Section 304.1 of the *International Mechanical Code*, shall not be required to comply with Sections 507.1.5, 507.2.3, 507.2.5, 507.2.8, 507.3.1, 507.3.3, 507.1.6 and 507.2.10 of the *International Mechanical Code*.
2. Factory-built commercial cooking recirculating systems that are *listed* and *labeled* in accordance with UL 710B, and installed in accordance with Section 304.1 of the *International Mechanical Code*, shall not be required to comply with Sections 507.1.5, 507.2.3, 507.2.5, 507.2.8, 507.3.1, 507.3.3, 507.1.6 and 507.2.10 of the *International Mechanical Code*. Spaces in which such systems are located shall be considered to be kitchens and shall be ventilated in accordance with Table 403.3.1.1 of the *International Mechanical Code*. For the purpose of determining the floor area required to be ventilated, each individual appliance shall be considered as occupying not less than 100 square feet (9.3 m^2).
3. Where cooking appliances are equipped with integral down-draft exhaust systems and such appliances and exhaust systems are *listed* and *labeled* for the application in accordance with NFPA 96, a hood shall not be required at or above them.
4. A Type I hood shall not be required for an electric cooking appliance where an *approved* testing agency provides documentation that the appliance effluent contains 5 mg/m^3 or less of grease when tested at an exhaust flow rate of 500 cfm (0.236 m^3/s) in accordance with UL 710B.

C A Type I hood (see definition of "Hood, Type I" in Section 202) is required for all appliances used for commercial cooking as defined in Section 202. In addition to the specific cooking appliances identified in the definition, further examples of commercial cooking appliances that require a commercial kitchen exhaust system are griddles (flat or grooved); tilting skillets or woks; braising and frying pans; roasters; pastry ovens; pizza ovens; charbroilers; salamanders and upright broilers; infrared broilers and open-burner stoves; ranges; and barbecue equipment. Further, the definition of "Commercial cooking appliances" defines a food service establishment as "...any building or portion thereof used for the preparation and serving of food."

The term "grease" refers to animal and vegetable fats and oils that are used to cook foods or that are a byproduct of cooking foods. Cooking appliances are primarily used for the preparation of food for compensation, trade or services rendered. Where the nature of the cooking produces grease or smoke, a Type I hood is required. The intent is not to require a Type I hood where there is a possibility of food being burned and producing smoke. For example, smoke that is produced when toast is burned does not mean that a Type I hood is required over a toaster.

Section 606 does not require exhaust hoods for cooking equipment or appliances installed outdoors where the grease-laden vapors, etc., discharge directly to the outside atmosphere, nor is this chapter intended to regulate cooking appliances installed in vehicles or towed trailers (see definition of "Commercial cooking appliances"). Note that cooking appliances installed outdoors but located under a roof should be evaluated for installation under a Type I hood just as if they were located inside a building having enclosing walls.

Exception 1 states the requirements for factory-built commercial exhaust hoods and specifies which code provisions are not applicable to factory-built hoods. Shop-built and field-constructed hoods are subject to all of the design and fabrication requirements of Section 507.2.

A factory-built commercial exhaust hood that has been tested in accordance with UL 710 and listed and labeled by an approved agency must be installed in accordance with the manufacturer's instructions. The importance of installing the system in strict compliance with the manufacturer's instructions cannot be overemphasized. These instructions contain specific installation requirements that are critical to the safe and efficient operation of the hood.

This exception states that listed and labeled hoods have demonstrated compliance with the construction and design requirements of this section and others, such as IMC Sections 507.1.5, 507.2.3, 507.2.5, 507.2.8, 507.3.1, 507.3.3, 507.1.6 and 507.2.10. Note that hoods listed and labeled in accordance with UL 710 are not exempt from Section 507.2.6; therefore, the 18-inch (457 mm) clearance requirement applies. However, Exception 2 of Section 507.2.6 of the IMC recognizes that some hoods can be listed and labeled for lesser clearances because UL 710 contains testing procedures for determining required clearances for hoods listed to that standard. The following is a list of some of the information that must be contained within the manufacturer's installation instructions or on the label:

- Minimum and maximum spacing between the front lower edge of the hood and the cooking surface.
- Minimum exhaust airflow quantity.
- Maximum supply airflow if the supply air is directed into the hood.
- Minimum overhangs of the exhaust hood over the cooking surface.
- Maximum allowable surface temperature of the cooking appliance.
- The specific type of cooking appliance an exhaust hood is intended to serve.

It is also important to determine that all parts and subassemblies of an exhaust hood are components of the listed and labeled exhaust hood, or that the parts and subassemblies have been evaluated under the same conditions of fire severity as the exhaust hood. Further, the exhaust hood must be compatible with and intended for the type of cooking appliance it will serve. The label of a factory-built hood tested in accordance with UL 710 will indicate the duty classification (e.g., extra-, heavy-, medium- and light-duty) of the appliances the hood is intended to serve. See Section 507.2.1 of the IMC. This will make it easy for the inspector in the field to determine whether the hood that is installed is suited for the appliances that are being installed under the hood.

It is not the intent of this section to require labeling of exhaust hoods. Unlabeled factory-built hoods and shop/field constructed hoods are permitted.

A factory-built or field-constructed commercial exhaust hood that has not been listed and labeled in accordance with UL 710 is permitted if it is designed, constructed and installed in accordance with Section 507 of the IMC and all other applicable requirements of this chapter. This section addresses hood material requirements, hood construction, hood dimensions, exhaust quantities and makeup air requirements. Kitchen exhaust systems must discharge all effluent to the outdoors to comply with Sections 501.2, 506.3.13 and 506.4 of the IMC.

Exception 2 allows installation of factory-built commercial cooking recirculating systems if they have been listed and labeled in accordance with UL 710B. It is important that recirculating systems be installed in accordance with the manufacturer's installation instructions so that the listing requirements are met. An improper installation could result in hazardous vapors being discharged back into the kitchen.

Recirculating systems are becoming more and more popular, especially in smaller businesses that serve food—for example, a sandwich shop that wants to expand its menu without becoming a full kitchen. A possible concern with using such a system is that they are tested to prevent particulate matter from being redistributed into the space, but they do not completely remove all contaminants and odors. For this reason, a space where one of these appliances is installed must be considered as a kitchen for the purpose of applying Table 403.3.1.1 of the IMC. A space not considered to be a traditional kitchen, and containing recirculating systems with no exhaust to the outdoors, would not be ventilated at all if not for this provision in the code. The ventilation rate for kitchens is 0.7 cfm/ft^2 of exhaust, so to apply this rate, each individual recirculating appliance is considered as occupying 100 square feet. The purpose of this requirement is to provide a minimal amount of exhaust in the area where one or more of these appliances is installed. If a recirculating appliance is installed in a typical commercial kitchen where the kitchen ventilation rate is being applied to the whole space, then the area of the kitchen would be used to determine the ventilation rate without adding the additional 100 square feet for each recirculating appliance. Note, however, if the kitchen is small and the number of recirculating appliances multiplied by 100 square feet is greater than the area of the kitchen, the exhaust rate for the kitchen must be based on the area calculated in accordance with this section.

All hoods, listed and unlisted, must capture and confine cooking vapors within the hood to prevent spillage into the room (see commentary, Section 507.1.6).

Commercial wood-fired baking ovens used for items such as pizza and bread are covered under UL 2162. The UL 2162 listing and labeling requirements must be followed for these ovens. Although venting beneath a Type I hood is not expressly required by the UL classification for these ovens, it is crucial to adhere to the manufacturer's instructions. Ventilation considerations are essential to appropriate combustion byproduct exhaustion and preserve interior air quality.

Exception 3 recognizes the hoodless griddle-type of cooking appliance that is becoming very popular in restaurants. These types of cooking appliances are often referred to as hibachi tables, where food is prepared for customers at the table where they are seated. These cooking tables have a built-in integral down-draft exhaust system that is designed to capture the cooking vapors by drawing the air across the table into exhaust inlets located at the edge of the cooking surface. The cooking vapors are routed to grease filters located under the cooking surface and then to the grease duct that is under the floor. The grease duct must be installed in accordance with IMC Section 506.3.10. These hoods must be listed and labeled for this application. Chapter 15 of NFPA 96 contains specific requirements for integral down-draft exhaust systems.

Exception 4 recognizes the growing use of small electrical appliances used for cooking, such as in small sandwich shops and convenience stores, where little or no grease is produced. The installation of a Type I hood in these small establishments would create the expense of the hood and the energy costs of running the fan and tempering the makeup air where grease emissions are minimal or nonexistent. The grease emission threshold requirement is consistent with NFPA 96 and the testing procedure is done in accordance with Section 17 of UL 710B. In order for an appliance to qualify for use without a Type I hood it must be tested by an approved agency and shown that the effluent contains 5 mg/m^3 or less of grease when tested at an exhaust flow rate of 500 cfm. See also the IMC definitions of “Light-,” “Medium-,” “Heavy-” and “Extra-heavy-duty cooking appliances” as well as IMC Section 507.2.

606.3 Operations and maintenance. Commercial cooking systems shall be operated, inspected and maintained in accordance with Sections 606.3.1 through 606.3.4.

C The provisions of this section introduce Sections 606.3.1 through 606.3.4, which contain requirements for controlling the hazards associated with the operation and maintenance of commercial cooking systems.

606.3.1 Ventilation system. The ventilation system in connection with hoods shall be operated at the required rate of air movement, and grease filters *listed* and *labeled* in accordance with UL 1046 shall be in place where equipment under a kitchen grease hood is used.

C The hood must be designed to adequately collect and exhaust fumes, smoke and vapors from the area over which it is installed. To accomplish this, the hood must cause an airflow pattern that will sweep and direct the fumes, smoke and vapors from the cooking surfaces into the hood inlet.

The IMC specifies the minimum quantity of exhaust air necessary for effective removal of cooking vapors and the approximate amount of makeup air necessary for proper operation. The quantity of required exhaust is as much a function of the operational characteristics of the cooking equipment as it is a function of the size of the cooking surface or the exhaust hood opening area and the presence of walls and side panels. Manufacturer recommendations must be followed where applicable.

Grease filters prevent large amounts of grease from collecting in the hood, in exhaust ducts, on fan blades and at the exhaust system termination. The accumulation of grease can cause blockage in ducts, cause equipment failure and create a fire hazard. It therefore makes sense to have grease filters in place whenever commercial cooking equipment is used. This section requires the grease filters to be listed and labeled in accordance with UL 1046. This is consistent with the requirements of the IMC.

606.3.2 Grease extractors. Where grease extractors are installed, they shall be operated when the commercial-type cooking equipment is used.

C As noted in the commentary for Section 606.3.1, it is imperative that grease removal devices operate where commercial cooking equipment is used. Grease extractors and similar grease removal devices range from simple designs, such as a configuration of baffles, to elaborate hot water scrubbers and electrostatic precipitators. The devices must be installed to comply with the IMC and the manufacturer’s installation instructions.

606.3.3 Cleaning. Hoods, grease-removal devices, fans, ducts and other appurtenances shall be cleaned at intervals as required by Sections 606.3.3.1 through 606.3.3.3.

C The provisions of this section introduce Sections 606.3.3.1 through 606.3.3.3, which contain requirements for controlling the hazards associated with grease accumulation in commercial cooking systems. This includes appliances that produce grease-laden vapors, the hood, any grease extractors, the exhaust duct and the exhaust fan. The date of the cleaning and the extent of the work performed must be documented and be maintained on the premises.

606.3.3.1 Inspection. Hoods, grease-removal devices, fans, ducts and other appurtenances shall be inspected at intervals specified in Table 606.3.3.1 or as *approved* by the *fire code official*. Inspections shall be completed by qualified individuals.

C A regular inspection schedule must be maintained to prevent the accumulation of grease residue within the exhaust system. The hood, grease removal devices, ducts, fans, fire suppression system discharge nozzles and other components of the system must be cleaned regularly to prevent excessive accumulation of grease. The frequency of such cleaning can vary, depending on the amount and type of usage; however, at a minimum, the equipment must be inspected at the intervals indicated in Table 606.3.3.1. Commercial cooking equipment exhaust systems may be viewed in some respects as a combustible material conveying system inasmuch as the grease carried within the duct is very combustible under the right conditions. As such, the inspection of the system should be entrusted only to persons trained and qualified in system inspection practices. While there are no identified nationally recognized criteria for gauging if an individual is or is not qualified to perform this work, several companies have developed their own certification programs for establishing requirements for individuals who supervise the cleaning of commercial cooking operations. In any case, the fire code official will make the final determination, in accordance with Section 102.8.

TABLE 606.3.3.1—COMMERCIAL COOKING SYSTEM INSPECTION FREQUENCY

TYPE OF COOKING OPERATIONS	FREQUENCY OF INSPECTION
High-volume cooking operations such as 24-hour cooking, charbroiling or wok cooking	3 months
Low-volume cooking operations such as places of religious worship, seasonal businesses and senior centers	12 months
Cooking operations utilizing solid fuel-burning cooking appliances	1 month
All other cooking operations	6 months

C Inspection frequencies are established in this table, which is based on the volume of cooking being performed, the type of cooking operation and the type of fuel used. The most restrictive requirement is for cooking operations using solid fuels, such as barbeque pits and meat smokers. When these or similar appliances use a solid fuel such as wood or charcoal, a minimum monthly inspection frequency is required. Cooking operations that use charbroilers or woks require a minimum 3-month inspection frequency, as do high volume cooking operations such as are found in 24-hour restaurants. The frequency of inspection is reduced to 12 months for lower intensity cooking activities including, but not limited to, seasonal businesses, places of worship and facilities for the care of senior citizens. All other cooking operations are subject to a 6-month inspection frequency. Depending on the nature of the commercial cooking activities, a single kitchen could have activities that require different inspection frequencies. For example, a restaurant that serves smoked meats would require that the appliance used for the preparation of the meat be inspected monthly, while other equipment would require inspection every 3 or 6 months, depending on the volume of cooking being performed.

606.3.3.2 Grease accumulation. If during the inspection it is found that hoods, grease-removal devices, fans, ducts or other appurtenances have an accumulation of grease, such components shall be cleaned in accordance with ANSI/IKECA C10.

C Frequent cleaning of exhaust systems is essential to the fire safety of cooking establishments to eliminate the potentially flammable and highly dangerous collection of grease within the exhaust system. The most common cleaning method is hand scraping, which requires that a person remove the hood filters and the access doors in the exhaust ductwork and physically scrape the accumulated grease from the interior surfaces of the exhaust system. Done properly, this method is effective but far from perfect. Regular inspection and diligence by the owner and the fire code official are needed to achieve the desired level of cleanliness of the exhaust system.

A frequently asked question about grease accumulation is, “How much of an accumulation is enough to warrant a cleaning?” Grease buildup in a commercial cooking exhaust system and the frequency with which the system must be cleaned depend on several factors, including but not limited to: the types of appliances under the hood, the type of grease-removal devices in use, exhaust velocity within the system, number of joints in the ductwork, changes in direction of the duct run, temperature of the grease-laden air being exhausted, efficiency of the exhaust fan and whether the duct is insulated or exposed to the outdoors. Just as the frequency of inspection in Table 606.3.3.1 varies based on the intensity of the cooking operation, so does the rate of grease accumulation. The referenced standard, ANSI/IKECA C10, provides requirements to determine the frequency and necessity for commercial kitchen exhaust system cleaning through inspection procedures. In addition, the standard defines acceptable methods for cleaning exhaust systems and components and sets standards for acceptable cleanliness.

606.3.3.3 Records. Records for inspections shall state the individual and company performing the inspection, a description of the inspection and when the inspection took place. Records for cleanings shall state the individual and company performing the cleaning and when the cleaning took place. Such records shall be completed after each inspection or cleaning and maintained.

C Adequate records of inspections and cleanings not only provide documentation of the required inspections and the resulting cleanings but are also essential in establishing a workable cleaning schedule. A record of all cleaning must be maintained by the person or party responsible for the system. The records must indicate the method of cleaning and the time between cleanings as well as what components were cleaned.

As a result of a review of Recommendation 2(c) of the NIST Charleston, South Carolina Sofa Superstore Fire Report, changes were made to Sections 110.2 and 110.3, along with 49 other sections (including this section), to comprehensively address recordkeeping requirements. Section 110.3 provides standardized recordkeeping requirements for periodic inspection, testing, servicing and other operational and maintenance requirements of the code and makes it clear that records must be maintained on the premises or another approved location and that copies of records must be provided to the fire code official upon request. Section 110.3 also makes it clear that records must be maintained for a period of not less than 3 years unless a different time interval is specified in the code or a referenced standard, and that the fire code official is authorized to prescribe the form and format of such records. See the commentary to Sections 110.2 and 110.3.

606.3.3.3.1 Tags. When a commercial kitchen hood or duct system is inspected, a tag containing the service provider name, address, telephone number and date of service shall be provided in a conspicuous location. Prior tags shall be covered or removed.

C This section details the markings required to visually confirm serviceability of commercial kitchen hood and duct systems. The text is consistent with the requirements set forth in ANSI/IKECA C10, as referenced in Section 606.3.3.2.

606.3.4 Extinguishing system service. *Automatic fire-extinguishing systems* protecting commercial cooking systems shall be serviced as required in Section 904.14.5.

C See the commentary to Section 904.14.5.

606.4 Appliance connection to building piping. Gas-fired commercial cooking appliances installed on casters and appliances that are moved for cleaning and sanitation purposes shall be connected to the piping system with an appliance connector *listed* as complying with ANSI Z21.69/CSA 6.16. The commercial cooking appliance connector installation shall be configured in accordance with the manufacturer's installation instructions. Movement of appliances with casters shall be limited by a restraining device installed in accordance with the connector and appliance manufacturer's instructions.

C This requirement is intended to end the practice of replacing listed flexible piping with residential flexible piping. Residential flexible piping is more easily damaged when the cooking equipment is moved for cleaning, thus causing a fire/life safety problem with gas leaks and fires. This section is also intended to limit the distance the appliances can be moved for cleaning to further protect the connection.

SECTION 607—COMMERCIAL COOKING OIL STORAGE

607.1 General. Storage of cooking oil (grease) in commercial cooking operations utilizing above-ground tanks with a capacity greater than 60 gal (227 L) installed within a building shall comply with Sections 607.2 through 607.7 and NFPA 30. For purposes of this section, cooking oil shall be classified as a Class IIIB liquid unless otherwise determined by testing.

C In these times of increasing interest in alternative fuels, used cooking oil has benefits in that it can be recycled for commercial cooking operations. It can also be chemically modified into biodiesel and used as fuel for mobile or stationary equipment. Because it can be recycled and reused, many restaurants and similar businesses that perform commercial cooking operations have found that capturing used cooking oil reduces waste disposal costs. As a result, the food service industry is seeking options for the safe storage of waste cooking oils in buildings and, as such, there has arisen a large market for collecting and recycling used cooking oil (grease) from commercial cooking operations. This is sometimes done using a system designed to store the used cooking oil on-site in an above-ground tank. These systems typically include wheeled recovery carts and hoses (see Commentary Figure 607.2) to assist in transferring the grease from the cooking appliance to the storage tank, and from the storage tank to a recovery truck. Some systems include heating elements that assist in keeping the grease in a form that is easily pumped to the recovery truck. This arrangement could create a problem if the system is not properly designed and installed. Without the requirements of Section 607 addressing the storage of cooking oils, many installations may not be installed with the safety features needed to protect employees and the public.

It should be noted that this section also addresses fresh cooking oil versus used, spent or inedible cooking oil. These oils need to be stored in tanks and related components that are food grade. That is why there are specific requirements for nonmetallic tanks. Most metallic tanks, as required by this section, could not meet these specifications.

This section also specifies that, for the application of these requirements, cooking oils are classified as Class IIIB liquids in accordance with the definition of "Combustible liquids" in Chapter 2 of the code. These liquids have a closed-cup-flash-point temperature greater than 200°F (93.33°C). Flash point and ignition temperatures for common cooking oils are shown in Commentary Figure 607.1 and the data confirms that this classification is correct.

Note that installation of the cooking oil storage tank and its piping will require a construction permit in accordance with Section 105.6.9.

Commentary Figure 607.1—Cooking Oil Flash Point and Ignition Temperatures

COOKING OIL TYPE	FLASH POINT TEMPERATURE (°F)	IGNITION TEMPERATURE (°F)
Canola oil	450	626
Corn oil	490	740
Cotton seed oil	486	650
Palm oil	323	600
Peanut oil	540	833
Soybean oil	549	833
Sunflower seed oil	550	Undetermined

For SI: °C = (°F - 32)/1.8.

607.2 Metallic storage tanks. Metallic cooking oil storage tanks shall be *listed* in accordance with UL 80 or UL 142, and shall be installed in accordance with the tank manufacturer's instructions.

C This section requires storage tanks for cooking oil storage to be listed as complying with either UL 142 or UL 80. Both standards are limited to shop-fabricated above-ground storage tanks (ASTs) designed to operate at atmospheric pressure. Both standards require tanks to be constructed of carbon steel meeting a certain specification and, before shipment, to be tested at the factory to confirm they are liquid tight. Tanks constructed to UL 80 have a maximum volume of 660 gallons (2271.25 L) versus UL 142, which does not limit the volume of ASTs. Installations of metallic ASTs for cooking oil storage also must comply with the manufacturer's installation instructions (see Commentary Figure 607.2).

Commentary Figure 607.2—Cooking Oil Storage Tank and Recovery Cart

Photo courtesy of Darling International Inc., Irving, TX

607.3 Nonmetallic storage tanks. Nonmetallic cooking oil storage tanks shall be *listed* in accordance with UL 2152 and shall be installed in accordance with the tank manufacturer's instructions. Tank capacity shall not exceed 200 gallons (757 L) per tank.

C As discussed in Section 607.1, cooking oil that has not been previously used must be stored in food grade tanks. Typically, nonmetallic tanks are more appropriate for unused cooking oil. These provisions are not limited to fresh cooking oil and can be utilized for used, spent and inedible cooking oils. This section references the appropriate standard, UL 2152, for such tanks. Essentially, a listing for use with cooking oil is required, as is making sure that the maximum temperature limits associated with the tank match the application in which the tank is used.

The tanks are limited in size to 200 gallons. Metallic tanks have the potential for much larger capacities, as discussed in the commentary for Section 607.2.

607.4 Cooking oil storage system components. Cooking oil storage system components shall include but are not limited to piping, connections, fittings, valves, tubing, hose, pumps, vents and other related components used for the transfer of cooking oil, and are permitted to be of either metallic or nonmetallic construction.

C This section simply describes what components make up a cooking oil storage system. Section 607.4.1 addresses the design and construction requirements for these components. This section also notes that these components are permitted to be either metallic or nonmetallic. The allowance for types of materials will depend on the requirements in Section 607.4.1.

607.4.1 Design standards. The design, fabrication and assembly of system components shall be suitable for the working pressures, temperatures and structural stresses to be encountered by the components.

C The performance-based language provided in this section requires that it be determined that the system's intended use is consistent with all the components of the system.

607.4.2 Components in contact with heated oil. System components that come in contact with heated cooking oil shall be rated for the maximum operating temperatures expected in the system.

C Cooking oil storage system components may be exposed to elevated temperatures, such as when draining spent oil from deep fat fryers. Manufacturers must document that components can withstand the highest temperature to be expected during normal operations to avoid component deterioration or failure from heat exposure.

607.5 Tank venting. Normal and emergency venting shall be provided for cooking oil storage tanks.

C This section and the UL tank standards referenced in Section 607.2 require ASTs to be equipped with a normal vent and an emergency vent.

607.5.1 Normal vents. Normal vents shall be located above the maximum normal liquid line, and shall have a minimum effective area not smaller than the largest filling or withdrawal connection. Normal vents shall be permitted to vent inside the building.

C Tanks are vented to maintain the internal tank pressure within the design operating range. Low pressure can increase the generation of vapors while high pressure can damage the tank or piping system. Any pressure outside of the design pressure range (which could be caused if the vent line were not located above the highest level of liquid) can have an adverse effect on the operation of the system as well as the piping and equipment. This section is also consistent with Section 21.4.3 of NFPA 30.

Recognizing the reduced hazard of Class IIIB liquids, this section does not require the normal vent for a cooking oil AST installed indoors to be terminated outside the building. Note, however, that Section 5704.2.7.3.3 of the code allows this condition only where the tank is equipped with a normally closed pressure/vacuum vent.

607.5.2 Emergency vents. Emergency relief vents shall be located above the maximum normal liquid line, and shall be in the form of a device or devices that will relieve excessive internal pressure caused by an exposure fire. For nonmetallic tanks, the emergency relief vent shall be allowed to be in the form of construction. Emergency vents shall be permitted to vent inside the building.

C All ASTs for cooking oil storage require an emergency vent that will relieve excessive internal pressure caused by exposure to fires in accordance with Section 5704.2.7.4; however, the emergency vents for ASTs storing Class IIIB liquids are generally permitted to discharge inside the building in accordance with that section.

607.6 Heating of cooking oil. Electrical equipment used for heating cooking oil in cooking oil storage systems shall be *listed* to UL 499 and shall comply with NFPA 70. Use of electrical immersion heaters shall be prohibited in nonmetallic tanks.

C Many cooking oil storage tank systems contain internal heaters to keep the oil above its melting temperature so that it can be efficiently removed by a recovery vacuum truck. This section requires that the electrical equipment associated with the heating of cooking oil be listed and its design and installation comply with NFPA 70. Note that this section does not require any temperature controls to ensure the cooking oil is not heated above its flash point temperature. See the commentary to Section 5701.5 for a discussion of the impact of heating combustible liquids above their flash point.

607.7 Electrical equipment. Electrical equipment used for the operation of cooking oil storage systems shall comply with NFPA 70.

C The phrase "cooking oil storage system" as used in this section could lead to the incorrect assumption that it is listed as a complete assembly of components. Fire code officials will need to evaluate the electrical equipment separately for compliance with the code and NFPA 70.

SECTION 608—MECHANICAL REFRIGERATION

[M] 608.1 Scope. Refrigeration systems shall comply with the *International Mechanical Code* and this section, as specified in Sections 608.1.1 and 608.1.2.

C IMC Chapter 11, in conjunction with ASHRAE 15 and IIAR 2, 3, 4 and 5 and IIAR CO2, provides complete coverage for the design, installation and maintenance of refrigeration systems.

608.1.1 Refrigerants other than ammonia. Refrigeration systems using a refrigerant other than ammonia shall comply with Section 608 and ASHRAE 15. Refrigeration systems containing carbon dioxide as the refrigerant shall also comply with IIAR CO2.

C This section is simply making sure that all refrigeration systems other than ammonia and the buildings that they are located in comply with ASHRAE 15 in addition to the requirements of Section 608. The standard IIAR CO2 for carbon dioxide refrigeration systems designed to be a companion to ASHRAE 15, providing additional design requirements that are unique to carbon dioxide systems. The standard is a supplement ASHRAE 15 that goes beyond the scope of ASHRAE 15 by regulating the complete lifecycle of carbon dioxide systems. Carbon dioxide has become increasingly popular as an industrial refrigerant because it is considered efficient and climate friendly.

608.1.2 Ammonia refrigeration. Refrigeration systems using ammonia refrigerant shall comply with IIAR 2 for system design; IIAR 6 for inspection, testing and maintenance; IIAR 7 for operating procedures; IIAR 8 for decommissioning; and IIAR 9 for engineering practices for existing systems. Refrigeration systems using ammonia refrigerant shall not be required to comply with Section 608.

C This section is specific to ammonia, whereas Section 608.1.1 deals with all other refrigerant types. The references in this section provide standards for ammonia refrigeration system design, maintenance, inspection and operating procedures and decommissioning. These standards are considered essential for facilities with ammonia refrigeration systems as a basis of providing for the safety of these facilities, as well as surrounding communities. Compliance with these standards fully addresses any hazards associated with ammonia and further application of Section 608 is not necessary.

608.2 Permits. An operational permit shall be obtained for refrigeration systems as set forth in Section 105.5.46.

C Only one system discussed in Chapter 6 requires permits: the operation of refrigeration systems. The permit for operation of refrigeration systems is intended to warn emergency responders that a potential hazard exists. This information will better equip them to respond to such a call (see commentary, Section 105).

[M] 608.3 Refrigerants. The use and purity of new, recovered and reclaimed refrigerants shall be in accordance with the *International Mechanical Code*.

C Refrigeration equipment is designed to operate with a specific type or types of refrigerant. Using the wrong refrigerant or a contaminated refrigerant could cause equipment damage or loss of operating efficiency. Refrigerant types differ in how they react with system lubricants, seals, gaskets and other components.

Existing equipment may have to be converted to a different type of refrigerant or charged with refrigerant recovered from it or another system; however, it would be an unnecessary risk to charge new equipment with any refrigerant other than that specified by the equipment manufacturer. See the commentary to IMC Section 1102.2.

[M] 608.4 Refrigerant classification. Refrigerants shall be classified in accordance with the *International Mechanical Code*.

C The classification of refrigerants is based on ASHRAE 34, which numbers and classifies refrigerants in accordance with their potential hazards. In the IMC, refrigerants are identified by their number, which is preceded by the letter "R" (for example, R-22). Trademark names of manufacturers, such as "Freon," are not used. The refrigerant number, name and chemical formula can be found in IMC Table 1103.1.

Each refrigerant is classified into a safety group that is based on two factors: flammability and toxicity. The safety group in which an individual refrigerant is classified relates to the potential hazard to building occupants and firefighters and is one of the factors used to determine the maximum quantities of refrigerants allowed by the IMC. See the commentary to IMC Section 1103.1.

[M] 608.5 Change in refrigerant type. A change in the type of refrigerant in a refrigeration system shall be in accordance with the *International Mechanical Code*.

C This section is intended to keep the local fire code official up-to-date on the status of large refrigeration systems and systems containing toxic and/or flammable refrigerants. See the commentary to IMC Section 1101.8.

608.6 Access. Access to refrigeration systems having a refrigerant circuit containing more than 220 pounds (100 kg) of Group A1 or 30 pounds (14 kg) of any other group refrigerant shall be provided for the fire department at all times as required by the *fire code official*.

C Where any one or more refrigeration circuits contain more than the specified quantity limits, the room or building housing the system or systems must be constructed with fire department access for emergency response, inspection and hazard assessment. Refrigerants of other than Group A1 tend to be more flammable or toxic. This section could require that the fire department be given keys to refrigeration machinery rooms and buildings.

608.7 Testing of equipment. Refrigeration equipment and systems having a refrigerant circuit containing more than 220 pounds (100 kg) of Group A1 or 30 pounds (14 kg) of any other group refrigerant shall be subject to periodic testing in accordance with Section 608.7.1. Records of tests shall be maintained. Tests of emergency devices or systems required by this chapter shall be conducted by persons trained and qualified in refrigeration systems.

C This section sets forth which refrigeration systems are subject to periodic testing and states that records of such testing must be maintained. Minimum qualifications are also required for testing personnel.

As a result of a review of Recommendation 2(c) of the NIST Charleston, South Carolina Sofa Superstore Fire Report, changes were made to Sections 110.2 and 110.3, along with 49 other sections (including this section), to comprehensively address recordkeeping requirements. Section 110.3 provides standardized recordkeeping requirements for periodic inspection, testing, servicing and other operational and maintenance requirements of the code and makes it clear that records must be maintained on the premises or another approved location and that copies of records must be provided to the fire code official upon request. Section 110.3 also makes it clear that records must be maintained for a period of not less than 3 years unless a different time interval is specified in the code or a referenced standard, and that the fire code official is authorized to prescribe the form and format of such records. See the commentaries to Sections 110.2 through 110.4 and 608.7.1.

608.7.1 Periodic testing. The following emergency devices or systems shall be periodically tested in accordance with the manufacturer's instructions and as required by the *fire code official*.

1. Treatment and flaring systems.
2. Valves and appurtenances necessary to the operation of emergency refrigeration control boxes.
3. Fans and associated equipment intended to operate emergency ventilation systems.
4. Detection and alarm systems.

C The devices and systems listed in this section are critical life safety and fire protection elements; therefore, it is imperative that they be tested periodically to assess their condition and dependability. Failure of the safety systems could lead to deadly consequences for building occupants and fire personnel entering the building in an emergency situation. See also the commentary to IMC Section 1109.1.

608.8 Emergency signs. Refrigeration units or systems having a refrigerant circuit containing more than 220 pounds (100 kg) of Group A1 or 30 pounds (14 kg) of any other group refrigerant shall be provided with *approved* emergency signs, charts and labels in accordance with NFPA 704. Hazard signs shall be in accordance with the *International Mechanical Code* for the classification of refrigerants listed therein.

C Signs, charts and labels are necessary to assist emergency response personnel in carrying out their duties to protect building occupants and protect themselves from the hazards associated with refrigerant chemicals (see IMC Table 1103.1).

608.9 Refrigerant detection. Machinery rooms shall be provided with a refrigerant detector with an audible and visible alarm. A detector, or a sampling tube that draws air to a detector, shall be provided at an *approved* location where refrigerant from a leak is expected to accumulate. The system shall be designed to initiate audible and visible alarms inside of and outside each entrance to the refrigerating machinery room and transmit a signal to an *approved* location where the concentration of refrigerant detected exceeds the lesser of the following:

1. The corresponding TLV-TWA values shown in the *International Mechanical Code* for the refrigerant classification.
2. Twenty-five percent of the lower flammable limit (LFL).

Detection of a refrigerant concentration exceeding the upper detection limit or 25 percent of the lower flammable limit (LFL), whichever is lower, shall stop refrigerant equipment in the machinery room in accordance with Section 608.10.1.

C This section triggers the requirement for refrigerant detection and alarms. IMC Section 1105.3 refers to this code for refrigerant detector requirements. Refrigerant detectors provide early warning of refrigerant leakage. Such leakage could result in a significant fire or health hazard if not discovered and stopped or if occupants are not evacuated from the building. Machinery rooms are required by the IMC where refrigerant quantities exceed specified limits.

Detector location is critical to early leakage warning and should comply with the detector manufacturer's instructions. The required detectors must be designed for application with the refrigerant or refrigerants used in the machinery room. Because machinery rooms are unattended most of the time, once the refrigerant gas is detected at the levels noted in this section (the lesser of the TLV-TWA values in the IMC or 25 percent of the LFL), a local alarm must be initiated and a signal must be transmitted to an approved location remote from the machinery room. Such locations typically include a security room or fire command center. A signal may also be transmitted to an on-duty, on-site technician via pager or cell phone.

The alarm is intended to alert those both inside the area of detection and in the immediate vicinity to prevent harm to those outside the area of refrigerant gas release. The notification to an approved location provides timely information to those who must take a role in emergency response, whereas the local alarm is a warning for those in the vicinity of the release. As a first step in the mitigation of the hazards of fugitive refrigerant gas, the required detectors have the additional important role of activating the emergency ventilation/exhaust systems in the machinery rooms required by IMC Section 1105.6.3 (see the definition of "Machinery room" and Table 1103.1, both in the IMC). This section also requires that the system be shut down in accordance with Section 608.10.1.

608.10 Remote controls. Where flammable refrigerants are used and compliance with Section 1106 of the *International Mechanical Code* is required, remote control of the mechanical equipment and appliances located in the machinery room as required by Sections 608.10.1 and 608.10.2 shall be provided at an *approved* location immediately outside the machinery room and adjacent to its principal entrance.

C Emergency controls located outside of the machinery room enclosure will allow shutdown of the compressors and related equipment without requiring someone to enter the room and risk being exposed to refrigerant or fire. This arrangement would also permit equipment shutdown by firefighting personnel without the risk of fire spreading into or out of the fire-resistance-rated enclosure. The controls must be located near the entrance to the machinery room so that their location is conspicuous. The controls should be labeled and color coded so that their purpose is obvious. Such controls are customarily painted red to make them readily identifiable as emergency devices.

608.10.1 Refrigeration system emergency shutoff. A clearly identified switch of the break-glass type or with an *approved* tamper-resistant cover shall provide off-only control of refrigerant compressors, refrigerant pumps and normally closed automatic refrigerant valves located in the machinery room. Additionally, this equipment shall be automatically shut off when the refrigerant vapor concentration in the machinery room exceeds the vapor detector's upper detection limit or 25 percent of the LEL, whichever is lower.

C This section of the code is intended to provide a safe environment for emergency response personnel when responding to an incident in a refrigeration room. Shutting down compressors and related refrigeration equipment could be necessary to prevent a hazardous condition from worsening and to allow the room to be safely entered. The emergency "kill" switch must be a tamper-resistant type (similar to manual fire alarm boxes) that requires more than one action to actuate it. To prevent an accidental startup, the switch must be capable of only stopping the controlled machinery, not restarting it. The switch must not affect the operation of life safety systems, such as detectors and exhaust equipment, and must not affect room and egress lighting. In addition to the manual switch, the required refrigerant room detector must also shut down the same equipment when the vapor concentration exceeds the lesser of the detector's upper detection limit or 25 percent of the refrigerant's lower explosive limit (LEL).

608.10.2 Ventilation system. A clearly identified switch of the break-glass type or with an *approved* tamper-resistant cover shall provide on-only control of the machinery room ventilation fans.

C For the same reasoning stated in the commentary to Section 608.10.1, a remote switch is required that is not connected to the exhaust system. Although not specifically stated, the intent is for the remote control to activate the emergency mode of operation (see commentary, IMC Section 1106.5.2). To maximize the dependability of the exhaust systems, ASHRAE 15 requires that they be powered from independent dedicated electrical branch circuits.

608.11 Emergency pressure control system. Permanently installed refrigeration systems in machinery rooms containing more than 6.6 pounds (3 kg) of flammable, toxic or highly toxic refrigerant or ammonia shall be provided with an emergency pressure control system in accordance with Sections 608.11.1 and 608.11.2.

C Technological advances in refrigeration system control equipment now make it possible to provide an automatic emergency control system to replace key functions of the traditional emergency control box. The automatic controls required by Sections 608.11.1 through 608.11.2.2 provide a means of mitigating an overpressure condition prior to operation of emergency pressure-relief vents and, most likely, prior to the arrival of emergency responders. The automatic valves also eliminate the need for emergency responders to decipher the condition of a system in an attempt to determine whether operation of manual crossover valves in an emergency control box would be of benefit in mitigating a system malfunction. The 6.6-pound (3 kg) threshold parallels provisions in Section 608.13.2. This section is referenced in IMC Section 1105.8 to increase the likelihood that designers don't miss this design requirement in the code. Note that these provisions are intended for permanently installed systems versus portable systems such as agricultural cooling trailers used in fields and at processing facilities. This is specifically clarified by the term "permanently installed."

Overall, these provisions add a requirement for a fully redundant safety control system in lieu of a manual system that has proven itself to be rarely, if ever, utilized by the fire service. This favorably resolves long-standing concerns regarding the potential for harm caused by an untrained person operating valves in an emergency control box. There is no condition where removal of refrigerant from a refrigeration system by the fire service is considered advisable. In contrast, automatic transfer of excess pressure to another zone of the system in conjunction with stopping the pressure source (i.e., the compressors) can safely mitigate an overpressure condition.

608.11.1 Automatic crossover valves. Each high- and intermediate-pressure zone in a refrigeration system shall be provided with a single automatic valve providing a crossover connection to a lower pressure zone. Automatic crossover valves shall comply with Sections 608.11.1.1 through 608.11.1.3.

C The requirement for a single crossover valve between systems is based on the traditional industry practice of providing a single manual crossover valve in the emergency control box. In the unlikely event that a fire causes an overpressure condition, allowing system zones to automatically interconnect creates a much larger heat sink to limit pressure buildup while safely containing refrigerant. If the exposure fire continues to grow, emergency relief vents can protect the refrigeration system and automatic reseating valves can automatically limit the release of refrigerant to the amount necessary to maintain the system within design limits. In contrast, most emergency responders would not possess the expertise to properly cycle a manual valve in an emergency control box to limit the release of refrigerant to the minimum amount necessary for safety.

608.11.1.1 Overpressure limit set point. Automatic crossover valves shall be arranged to automatically relieve excess system pressure to a lower pressure zone if the pressure in a high- or intermediate-pressure zone rises to within 90 percent of the set point for emergency pressure relief devices.

C This section provides a safety buffer between activation of the emergency pressure control system (EPCS) and operation of a relief valve. Because of variances in operational tolerances among relief valves and because some relief valves may begin to seep at 90 percent of their rated operating pressure, it is appropriate to have the EPCS shut down a system if system pressure rises to 90 percent of the relief valve set pressure. This further reduces the potential for any release from a system that has malfunctioned and overpressurized.

608.11.1.2 Manual operation. Where required by the *fire code official*, automatic crossover valves shall be capable of manual operation.

C This section authorizes the local fire code official to require manual control capabilities for the crossover valve. Although this provision is not regarded as necessary by the industry, it is recognized that some fire departments will prefer to have the manual control capability for backup.

608.11.1.3 System design pressure. Refrigeration system zones that are connected to a higher pressure zone by an automatic crossover valve shall be designed to safely contain the maximum pressure that can be achieved by interconnection of the two zones.

C This provision requires lower pressure zones to be capable of handling additional pressure added by a crossover condition without overpressurizing or operating the emergency relief vents on the lower zone. The legacy codes did not address this concern, given the assumption that someone operating a manual bypass valve in the emergency control box would be knowledgeable with regard to system limitations; however, this may or may not be true. Nevertheless, the legacy codes never required the low pressure side of the system to handle the high-side pressure, and as a result, some systems with emergency control boxes present the potential for an emergency responder to overpressurize a system zone by fully opening a manual crossover valve too quickly. The resulting overpressure condition could cause operation of a relief vent or even a failure in the piping system.

608.11.2 Automatic emergency stop. An automatic emergency stop feature shall be provided in accordance with Sections 608.11.2.1 and 608.11.2.2.

C This establishes applicability of the requirements for the additional safety feature of an emergency stop feature in refrigeration systems.

608.11.2.1 Operation of an automatic crossover valve. Operation of an automatic crossover valve shall cause all compressors on the affected system to immediately stop. Dedicated pressure-sensing devices located immediately adjacent to crossover valves shall be permitted as a means for determining operation of a valve. To ensure that the automatic crossover valve system provides a redundant means of stopping compressors in an overpressure condition, high-pressure cutout sensors associated with compressors shall not be used as a basis for determining operation of a crossover valve.

C This section is intended to ensure the automatic crossover system has a fully redundant means of stopping compressors. Compressors are ordinarily provided with automatic high-pressure cutout controls, but this section requires that these controls not be used to satisfy the requirement. An additional set of controls can be required to serve as a backup means of preventing a severe overpressure condition that could cause operation of an emergency relief vent.

608.11.2.2 Overpressure in low-pressure zone. The lowest pressure zone in a refrigeration system shall be provided with a dedicated means of determining a rise in system pressure to within 90 percent of the set point for emergency pressure relief devices. Activation of the overpressure sensing device shall cause all compressors on the affected system to immediately stop.

C The lowest pressure zone of a system cannot be arranged to bleed pressure to another system zone since crossing the lowest pressure zone to a higher pressure zone would most likely result in reverse flow. However, by providing a redundant emergency stop control, which would disengage the compressor, an overpressure condition should be automatically mitigated. Overpressure on a low-pressure zone would most likely result from a defrost line from the high side that is stuck in the open position, and stopping the compressor will disengage the pressure source for the defrost system. Note that compressors will only cut out if an overpressure condition occurs. If the emergency condition involves a leak on the low side, compressors will continue to operate, which is beneficial in pumping down the low side for this type of event.

608.12 Storage, use and handling. Flammable and combustible materials shall not be stored in machinery rooms for refrigeration systems having a refrigerant circuit containing more than 220 pounds (100 kg) of Group A1 or 30 pounds (14 kg) of any other group refrigerant. Storage, use or handling of extra refrigerant or refrigerant oils shall be as required by Chapters 50, 53, 55 and 57.

Exceptions: This provision shall not apply to:

1. Spare parts, tools and incidental materials necessary for the safe and proper operation and maintenance of the system.
2. Refrigerant removed from equipment during a repair or replacement and temporarily stored in a pressure vessel complying with ASME BPVC Section VIII, for reuse after the repair or replacement has been completed.

C Storage of materials could introduce additional hazards in rooms already considered to be hazardous because of large quantities of refrigerants in the system circuits and machines. Exception 1 recognizes that certain items such as spare parts and associated tools related to operation and maintenance should be allowed in such rooms. Exception 2 recognizes the reality that during repairs and replacement refrigerant often needs to be removed from a system. The machinery room has refrigerant detectors as well as ventilation in the event of a leak. Hence, the machinery room is an appropriate environment for temporarily storing the refrigerant that will be added back into the system following any repair or replacement. Since the refrigerant is either in the refrigeration equipment or pressure vessel, there is no added hazard to the machinery room. It should be noted that in a machinery room any group of refrigerants can be used in the refrigeration equipment.

608.13 Discharge and termination of pressure relief and purge systems. Pressure relief devices, fusible plugs and purge systems discharging to the atmosphere from refrigeration systems containing flammable, toxic or highly toxic refrigerants or ammonia shall comply with Sections 608.13.2 and 608.13.3.

C Discharge systems are intended to treat, incinerate or absorb flammable or toxic refrigerants that would otherwise be released unaltered into the atmosphere. Release would result from the operation of pressure relief devices or intentional dumping of refrigerant in an emergency. Specific requirements are found in the subsections for each type of refrigeration system.

608.13.1 Fusible plugs and rupture members. Discharge piping and devices connected to the discharge side of a fusible plug or rupture member shall have provisions to prevent plugging the pipe in the event the fusible plug or rupture member functions.

C This section is intended to ensure that, through the use of discharge piping and devices, the emergency pressure relief system will not become obstructed in the event that a fusible plug or rupture member operates and causes debris to be ejected into the relief system.

608.13.2 Flammable refrigerants. Systems containing more than 6.6 pounds (3 kg) of flammable refrigerants having a density equal to or greater than the density of air shall discharge vapor to the atmosphere only through an *approved* treatment system in accordance with Section 608.13.4 or a flaring system in accordance with Section 608.13.5. Systems containing more than 6.6 pounds (3 kg) of flammable refrigerants having a density less than the density of air shall be permitted to discharge vapor to the atmosphere provided that the point of discharge is located outside of the structure at not less than 15 feet (4572 mm) above the adjoining grade level and not less than 20 feet (6096 mm) from any window, ventilation opening or *exit*.

C Where they have a density greater than air (i.e., are heavier than air) and pose the hazard of collecting in low points, which could bring them into contact with ignition sources, flammable refrigerants (A2, B2, A3, B3) must be incinerated in a flaring system (see Section 6004.2.2.7.1). The second sentence of the provision is derived from ANSI/ASHRAE 15 and recognizes the reduced hazard of lighter-than-air flammable refrigerants by allowing them to discharge to the atmosphere without incineration or treatment since they would either dissipate into the air or flare at the point of discharge. Because of their flammability hazard, however, the point of discharge must be located out of reach from grade and well away from building openings. Note that these provisions apply only to systems that contain more than 6.6 pounds (3 kg) of refrigerant because systems containing smaller quantities are considered "small systems" and their relative hazard is considered insignificant.

608.13.3 Toxic and highly toxic refrigerants. Systems containing more than 6.6 pounds (3 kg) of toxic or highly toxic refrigerants shall discharge vapor to the atmosphere only through an *approved* treatment system in accordance with Section 608.13.4 or a flaring system in accordance with Section 608.13.5.

C Toxic refrigerants, like flammable refrigerants, must be treated to reduce their toxicity or be managed through a treatment system or destroyed by incineration (see Section 6004.2.2.7.1).

608.13.4 Treatment systems. Treatment systems shall be designed to reduce the allowable discharge concentration of the refrigerant gas to not more than 50 percent of the IDLH at the point of exhaust. Treatment systems shall be in accordance with Chapter 60.

C See the commentary to Section 6004.2.2.7. "Immediately Dangerous to Life and Health (IDLH)" is defined in Section 202.

608.13.5 Flaring systems. Flaring systems for incineration of flammable refrigerants shall be designed to incinerate the entire discharge. The products of refrigerant incineration shall not pose health or environmental hazards. Incineration shall be automatic upon initiation of discharge, shall be designed to prevent blowback and shall not expose structures or materials to threat of fire. Standby fuel, such as LP-gas, and standby power shall have the capacity to operate for one and one-half the required time for complete incineration of refrigerant in the system. Standby electrical power, where required to complete the incineration process, shall be in accordance with Section 1203.

C Destruction of refrigerant by incineration is supposed to render the discharge harmless, so obviously the flames and the combustion byproducts do not pose a hazard themselves. Because most refrigerants would not support combustion unaided, a fuel source is necessary to sustain incineration.

608.14 Mechanical ventilation exhaust. Exhaust from mechanical ventilation systems serving refrigeration machinery rooms containing flammable, toxic or highly toxic refrigerants capable of exceeding 25 percent of the LFL or 50 percent of the IDLH shall be equipped with *approved* treatment systems to reduce the discharge concentrations to those values or lower.

Exception: Refrigeration systems containing Group A2L complying with Section 608.18.

C This section is merely a trigger for the need of a treatment system when there is the possibility that exhaust from such systems is either flammable or toxic.

The exception to this section recognizes the reduced risk of the lower flammability rating of A2L refrigerants when they comply with Section 608.18. Essentially, Section 608.18 provides detection and ventilation, which removes the need to provide a treatment system as the flammability concentration is much lower.

608.15 Notification of refrigerant discharges. The *fire code official* shall be notified immediately when a discharge becomes reportable under state, federal or local regulations in accordance with Section 5003.3.1.

C Emergency personnel must be informed of a discharge so that they can respond appropriately. The refrigerant discharge notification requirements of this section parallel those required for all hazardous materials in Section 5003.3.1. There is no reason for different requirements for refrigerants than for other hazardous materials.

608.16 Records. A record of refrigerant quantities brought into and removed from the premises shall be maintained.

C Emergency personnel must be able to maintain accurate assessments of the dangers they may face and the hazards the public may face in and around buildings housing refrigeration systems.

As a result of a review of Recommendation 2(c) of the NIST Charleston, South Carolina Sofa Superstore Fire Report, changes were made to Sections 110.2 and 110.3, to comprehensively address record-keeping requirements. Section 110.3 provides standardized record-keeping requirements for periodic inspection, testing, servicing and other operational and maintenance requirements of the code and makes it clear that records must be maintained on the premises or another approved location and that copies of records must be provided to the fire code official upon request. Section 110.3 also makes it clear that records must be maintained for a period of not less than 3 years unless a different time interval is specified in the code or a referenced standard, and that the fire code official is authorized to prescribe the form and format of such records. See the commentaries to Sections 110.2 and 110.3.

[M] 608.17 Electrical equipment. Where refrigerant of Groups A2, A3, B2 and B3, as defined in the *International Mechanical Code*, are used, refrigeration machinery rooms shall conform to the Class I, Division 2, hazardous location classification requirements of NFPA 70.

C This section mirrors the text of IMC Section 1106.3 and is included in the code because, in some jurisdictions, the fire code official is designated to inspect classified electrical equipment. A reference to classified electrical requirements here is consistent with the approach taken in Chapters 50 and 57 for other hazardous materials and flammable liquids, and it provides the fire code official with the provisions that are to be enforced.

[M] 608.18 Group A2L and B2L refrigerant. Machinery rooms for Group A2L and B2L refrigerant shall comply with Sections 1106.4.1 through 1106.4.3 of the *International Mechanical Code*.

C Group A2L and B2L refrigerant machinery rooms must adhere to IMC Sections 1106.4.1 through 1106.4.3. They are HFC or HFO chemicals, or mixes thereof, and are described as “mildly flammable.” The minimum ignition energy and burning velocity of A2L refrigerants are higher. To achieve 0 percent ozone depletion potential, a GWP of as near to 1 (CO_2) as possible, fire safety and refrigeration cycle efficiency, refrigerant chemical compositions have advanced from CFCs to HCFCs to HFCs and finally to HFOs. When released into the earth's atmosphere, HFO refrigerants have a minimal atmospheric lifespan compared to many other substances that can survive hundreds or thousands of years.

608.18.1 Elevated temperatures. Open flame-producing devices or continuously operating hot surfaces over 1,290°F (700°C) shall not be permanently installed in the room.

C The machinery room prohibits permanent installations of open flame-producing equipment or hot surfaces that run continuously over 1,290°F (700°C). This restriction aims to avoid the possible fire risks associated with high temperatures and open flame-producing devices.

[M] 608.18.2 Refrigerant detector. In addition to the requirements of Section 1105.3 of the *International Mechanical Code*, refrigerant detectors shall signal an alarm and activate the ventilation system in accordance with the response time specified in Table 608.18.2.

C Refrigerant detectors must trigger an alert and turn on the ventilation system in accordance with Table 608.18.2 response time requirements. Once turned on, the emergency ventilation system cannot be turned off automatically. Table 608.18.2

lists the ventilation rates corresponding to particular A2L refrigerants. It should be noted that the table gives CFM exhaust rates without accounting for the machinery room's size or the system's refrigerant charge in pounds. Note that Table 608.18.2 can be substituted with the requirements of ASHRAE 15.

TABLE 608.18.2—GROUP A2L AND B2L DETECTOR ACTIVATION

ACTIVATION LEVEL	MAXIMUM RESPONSE TIME (seconds)	ASHRAE 15 VENTILATION LEVEL	ALARM RESET	ALARM TYPE
Less than or equal to the OEL in Table 1103.1 of the *International Mechanical Code*	300	1	Automatic	Trouble
Less than or equal to the refrigerant concentration level in Table 1103.1 of the *International Mechanical Code*	15	2	Manual	Emergency

C Ensuring adequate ventilation and air quality in different occupancy types requires meeting the minimum exhaust rates mentioned in Table 608.18.2 for refrigerants. The exhaust system minimum airflow rates are calculated using the table criteria. These rates are essential for ridding indoor environments of pollutants, smells and excess heat. Different refrigerant types have different exhaust rates.

[M] 608.18.3 Mechanical ventilation. The machinery room shall have a mechanical ventilation system complying with ASHRAE 15.

C The mechanical ventilation system in the equipment room shall be ASHRAE 15 compliant. Refrigerant is expected in the mechanical ventilation exhaust because refrigerant detectors activate the exhaust. The exhaust output must be directed considerably away from people and from openings that could allow refrigerant to reenter the building since the combination of air and refrigerant could be in a dangerous concentration.

SECTION 609—HYPERBARIC FACILITIES

609.1 General. Hyperbaric facilities shall be inspected, tested and maintained in accordance with NFPA 99.

C IBC Section 425 requires that hyperbaric facilities, regardless of occupancy, comply with NFPA 99 for installation. This section provides the maintenance requirements so that such facilities continue to operate safely. The standard provides detailed responsibilities for testing and inspecting these facilities, such as requirements for an on-site hyperbaric safety director and for training.

609.2 Records. Records shall be maintained of all testing and repair conducted on the hyperbaric chamber and associated devices and equipment. Records shall be available to the *fire code official*.

C This section requires that records be kept to document compliance with the maintenance and testing requirements of NFPA 99 for the benefit of the facility owner and the fire code official.

As a result of a review of Recommendation 2(c) of the NIST Charleston, South Carolina Sofa Superstore Fire Report, changes were made to Sections 110.2 and 110.3, along with 49 other sections, to comprehensively address recordkeeping requirements. Section 110.3 provides standardized recordkeeping requirements for periodic inspection, testing, servicing and other operational and maintenance requirements of the code and makes it clear that records must be maintained on the premises or another approved location and that copies of records must be provided to the fire code official upon request. Section 110.3 makes it clear that records must be maintained for a period of not less than 3 years unless a different time interval is specified in the code or a referenced standard, and that the fire code official is authorized to prescribe the form and format of such records. See the commentaries to Sections 110.2 and 110.3.

SECTION 610—CLOTHES DRYER EXHAUST SYSTEMS

610.1 Clothes dryer exhaust duct systems. Clothes dryer exhaust duct systems shall be in accordance with Sections 610.1.1 and 610.1.2.

C This section references the following subsections for clothes dryer exhaust duct system requirements. The term “exhaust duct system” is consistent with the terms used in the IFGC, IMC and *International Residential Code*® (IRC®). These systems include the termination outlet and may include dryer exhaust duct power ventilators, which are also known as booster fans. The IFGC and IMC both include a definition for “Clothes dryer” in Chapter 2. The IFGC definition includes sub-definitions for Type 1 and Type 2 dryers.

610.1.1 Installation. Clothes dryer exhaust duct systems shall be installed in accordance with the *International Mechanical Code* or the *International Fuel Gas Code*, and the manufacturer's installation instructions.

C IMC Section 504 and IFGC Section 614 provide specific requirements for clothes dryer exhaust systems. The IFGC would apply only to dryers that are fuel gas appliances, as stated in the scope of IFGC (see IFGC Section 101.2). In addition to these requirements, the installation is to comply with the manufacturer's installation instructions.

610.1.2 Maintenance. The lint trap, mechanical and heating components, and the exhaust duct system of a clothes dryer shall be maintained in accordance with the manufacturer's operating instructions to prevent the accumulation of lint or debris that prevents the exhaust of air and products of combustion.

C There are many statistics published each year regarding clothes dryer fires, such as CPSC, NFPA and USFA. There are approximately 15,600 structure fires, 400 injuries and 15 deaths reported annually as a result of dryer fires. According to the United States Fire Administration, every year clothes dryer fires account for over $100 million in losses. Also, dryer fires involving commercial dryers have a 78 percent higher injury rate than residential dryer fires. A majority of dryer fires occur as a result of highly flammable lint getting caught in the dryer's vent and becoming heated to the point of ignition. While many of the statistics address residential applications, there are also some statistics that identify issues in commercial applications, too.

Maintaining clothes dryers and clothes dryer exhaust duct systems in any occupancy is critical to reducing the fire hazard. The IFGC, IMC and IRC require commercial and residential clothes dryers to be listed and labeled and to be installed in accordance with the manufacturer's installation instructions. The required product testing standards include requirements for specific cleaning and maintenance directions to be part of the manufacturer's installation and use instructions. The frequency of inspections depends on various factors, such as how often the dryer is used, the geometry of the exhaust duct system, and the age and type of dryer.

Bibliography

The following resource materials were used in the preparation of the commentary for this chapter of the code:

Commercial Kitchen Hood—Application Guide. Washington, DC: International Code Council, 2010.

IRC—24, *International Residential Code*. Washington, DC: International Code Council, 2024.

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